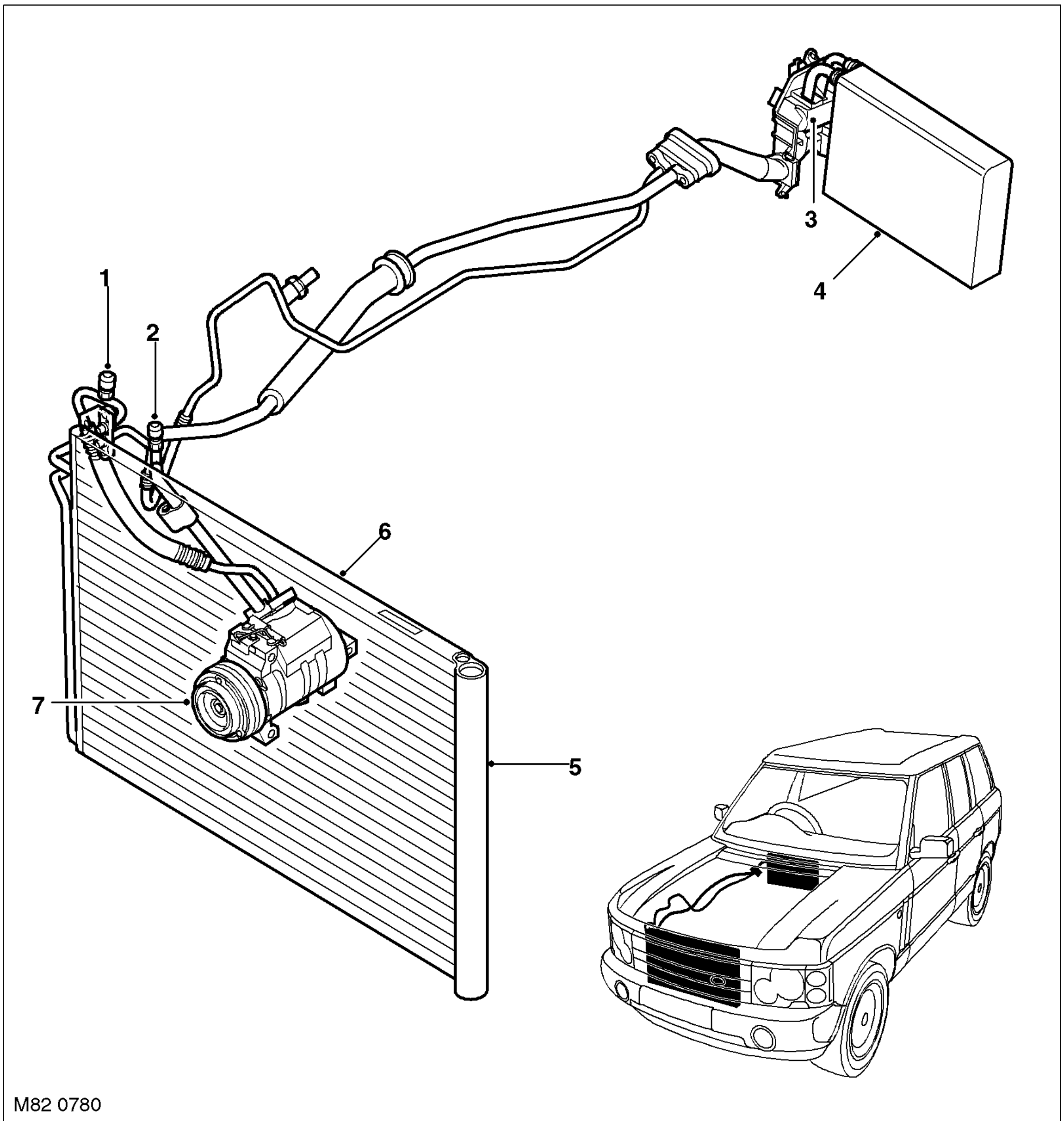
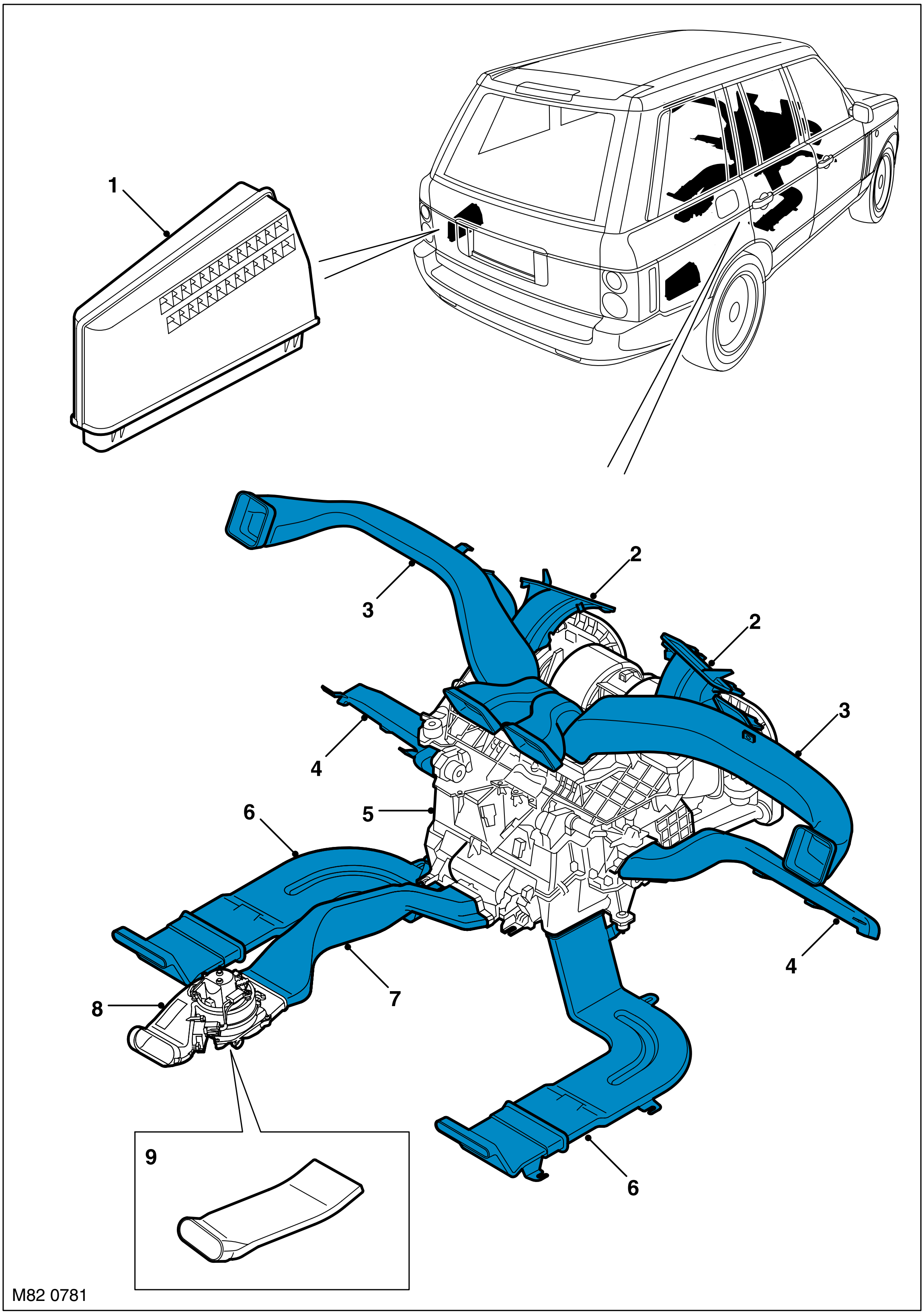
Refrigerant System Component Location



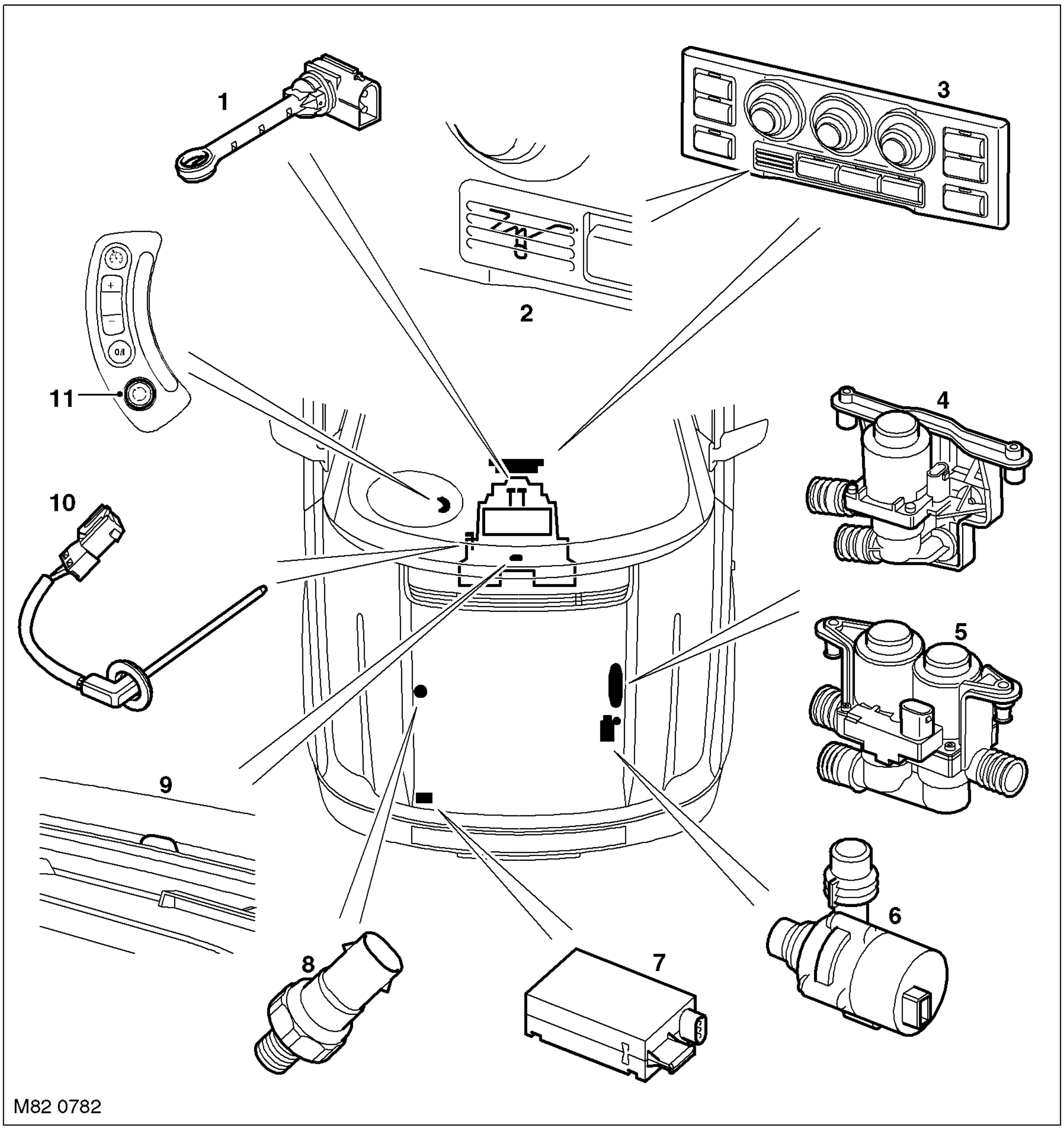
1 High pressure servicing connection
2 Low pressure servicing connection
3 Thermostatic expansion valve
4 Evaporator
5 Receiver drier module
6 Condenser
7 Compressor

Heater Assembly and Ducting Component Location



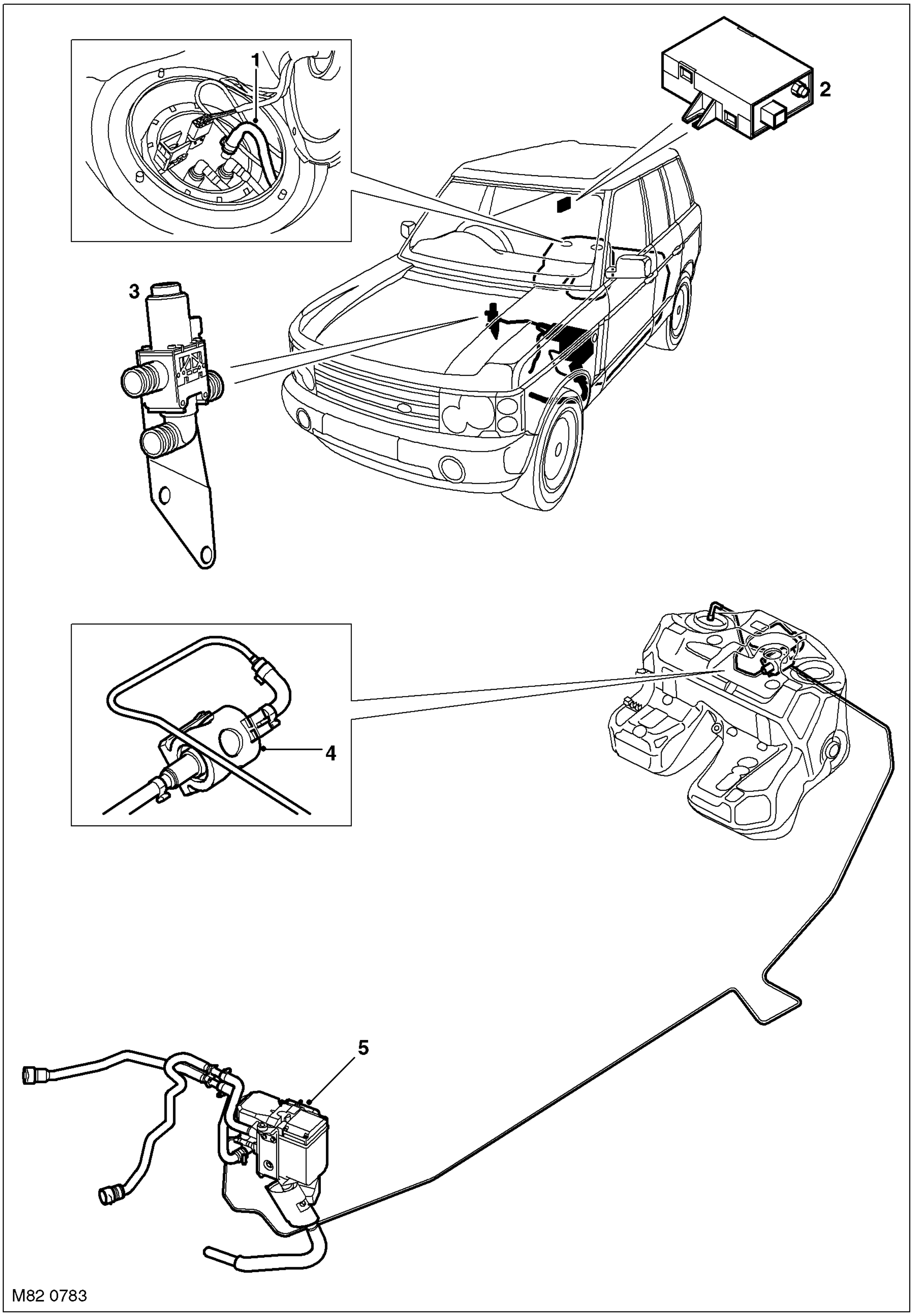
1 Forced ventilation outlet
2 Windscreen duct
3 Face level duct
4 Front footwell duct
5 Heater assembly
6 Rear footwell duct
7 Rear face duct
8 Rear blower (high line system only)
9 Rear face duct extension (low line system only)

Control System Component Location



- **1** Heater temperature sensor (LH on low line, LH & RH on high line)
- **2** In-car temperature sensor
- **3** ATC ECU (high line version)
- **4** Single coolant valve (low line only)
- **5** Dual coolant valve (high line only)
- **6** Auxiliary coolant pump
- **7** Pollution sensor (high line only)
- **8** Refrigerant pressure sensor
- **9** Sunlight sensor (high line only)
- **10** Evaporator temperature sensor
- **11** Steering wheel recirculation switch

FBH System Component Location



1 FBH fuel pipe tank connection
2 FBH receiver (where fitted)
3 Changeover valve (where fitted)
4 FBH fuel pump
5 FBH unit

Description

General
Either a low or high line Air Conditioning (A/C) system is installed on the vehicle. The low line system features automatic control of a common temperature setting for both sides of the passenger compartment, with manual control of the blower speed, air recirculation and air distribution. The high line system is fully automatic with separate temperature settings for the LH and RH sides of the passenger compartment and manual overrides for blower speed, air recirculation and air distribution. The high line system also incorporates independent temperature and volume control for the rear passengers. The low and high line systems both include:

- A refrigerant system
- An air inlet housing
- A heater assembly
- Distribution ducts
- Two forced ventilation outlets
- An Automatic Temperature Control (ATC) ECU and sensors
- An auxiliary coolant pump
- A coolant valve.

The high line system incorporates the following additional components:

- A second coolant valve
- A pollution sensor
- A rear blower unit.

A Fuel Burning Heater (FBH) system is installed as standard on Td6 models and as an option on V8 models. The FBH is a supplementary heater installed in the heater coolant circuit. The FBH can be operated without the engine running to pre-heat the passenger compartment and after the engine starts to reduce the heater coolant warm-up time. On Td6 models, the FBH also boosts the heater coolant temperature while the engine is running at low ambient air and engine coolant temperatures, to maintain heater performance at an acceptable level.

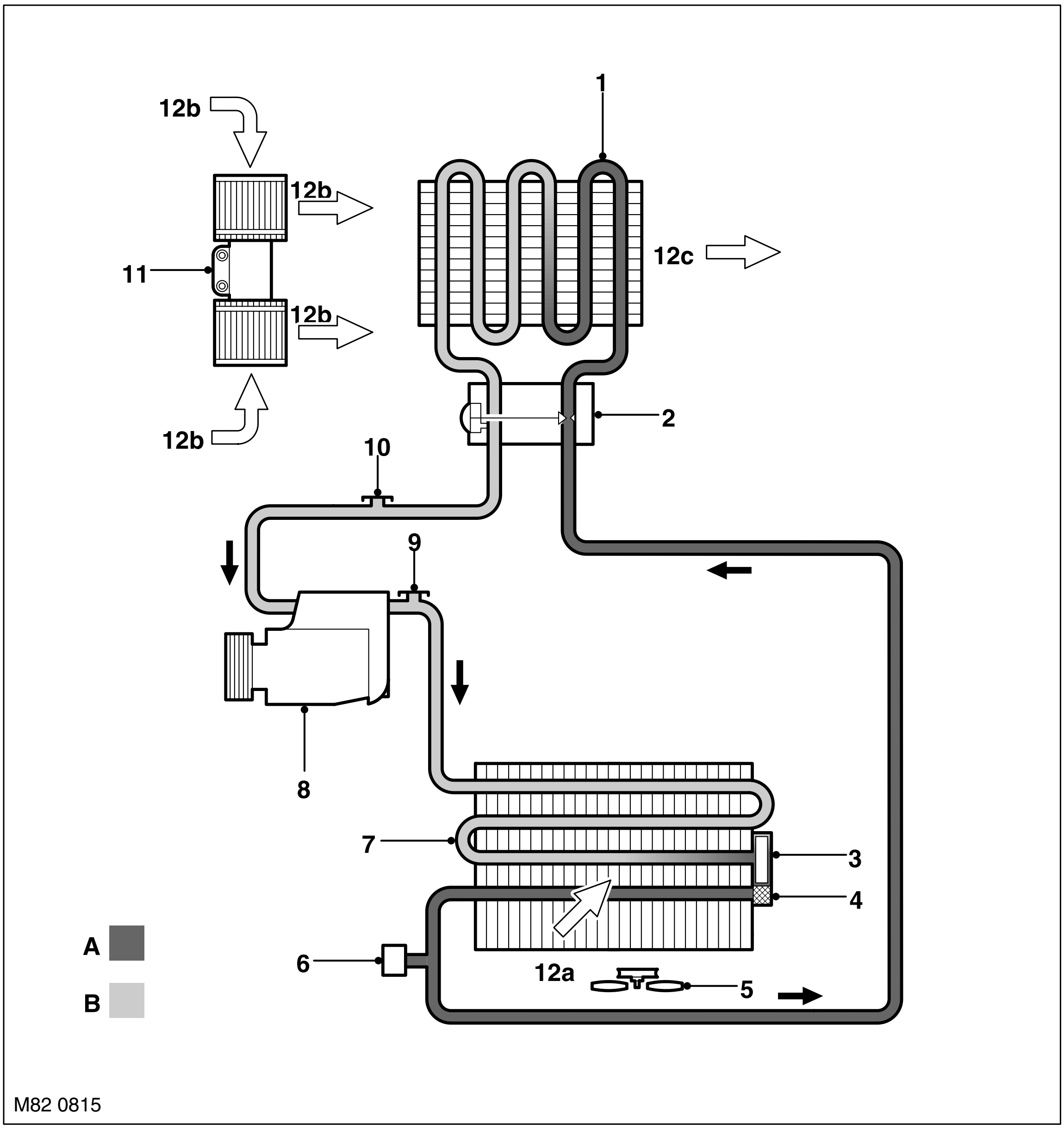
Refrigerant System
The refrigerant system transfers heat from the vehicle interior to the outside atmosphere to provide the heater assembly with dehumidified cool air. The system comprises:

- A compressor
- A condenser and receiver drier
- A thermostatic expansion valve
- An evaporator
- Refrigerant lines.

The system is a sealed, closed loop, filled with a charge weight of R134a refrigerant as the heat transfer medium. Oil is added to the refrigerant to lubricate the internal components of the compressor.

To accomplish the transfer of heat, the refrigerant is circulated around the system, where it passes through two pressure/temperature regimes. In each of the pressure/temperature regimes, the refrigerant changes state, during which process maximum heat absorption or release occurs. The low pressure/temperature regime is from the thermostatic expansion valve, through the evaporator to the compressor; the refrigerant decreases in pressure and temperature at the thermostatic expansion valve, then changes state from liquid to vapour in the evaporator, to absorb heat. The high pressure/temperature regime is from the compressor, through the condenser and receiver drier to the thermostatic expansion valve; the refrigerant increases in pressure and temperature as it passes through the compressor, then releases heat and changes state from vapour to liquid in the condenser.

Refrigerant System Schematic



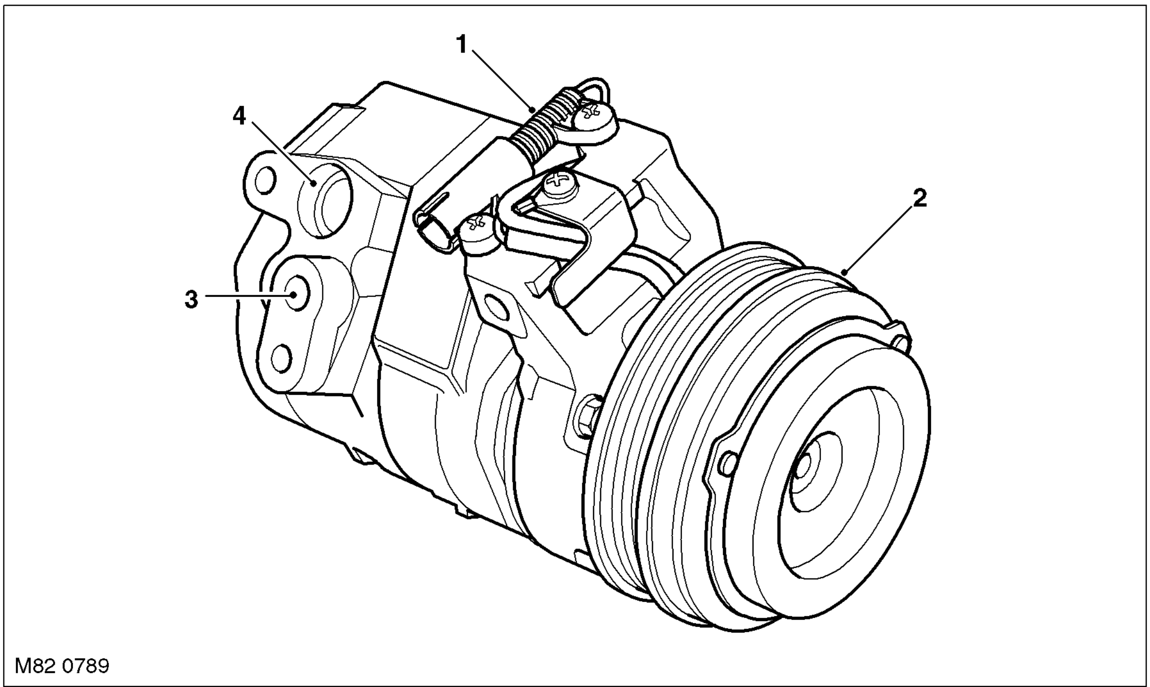
1 Evaporator
2 Thermostatic expansion valve
3 Desiccant (in modulator)
4 Filter (in modulator)
5 Electric cooling fan
6 Refrigerant pressure sensor
7 Condenser
8 Compressor
9 High pressure servicing connection
10 Low pressure servicing
11 Blower
12 Air flows:
- **a** Ambient air flow through condenser
- **b** Fresh/Recirculated air flow through blower
- **c** Cooled air flow to vehicle interior

Compressor

The compressor circulates the refrigerant around the system by compressing low pressure, low temperature vapour from the evaporator and discharging the resultant high pressure, high temperature vapour to the condenser.

The compressor is a fixed displacement unit attached to a mounting bracket on the engine. A dedicated drive belt on the engine crankshaft drives the compressor via a pulley and an electrically actuated clutch. Operation of the clutch is controlled by the ATC ECU.

Compressor Assembly



1 Clutch connector
2 Pulley
3 Outlet port
4 Inlet port

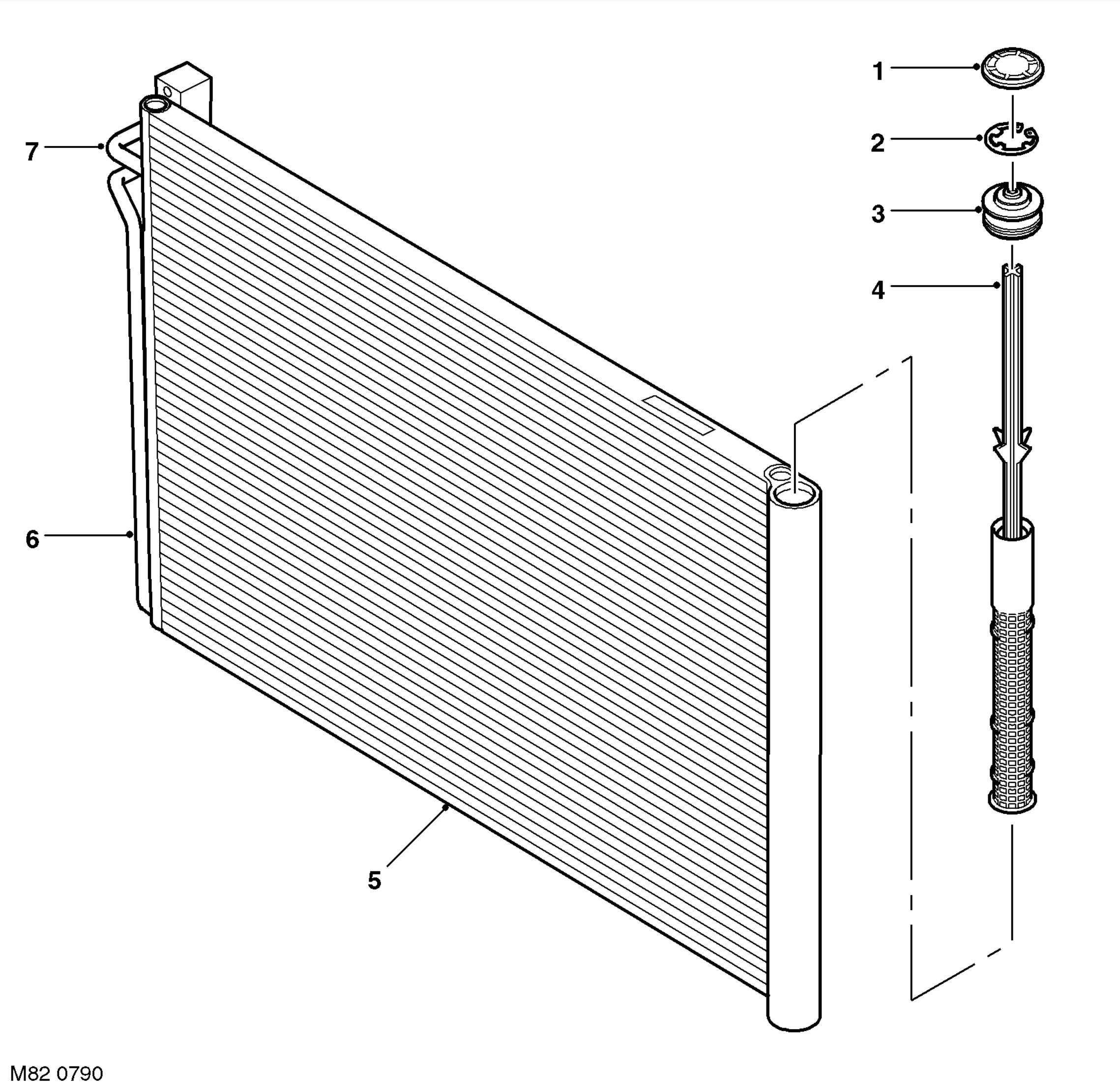
Condenser

The condenser transfers heat from the refrigerant to the surrounding air to convert the vapour from the compressor into a liquid. A receiver drier module, integrated onto the LH side of the condenser, incorporates a filter and a desiccant to remove solid impurities and moisture from the refrigerant. The receiver drier module also functions as a reservoir for liquid refrigerant, to accommodate changes of heat load at the evaporator.

The condenser is installed immediately in front of the radiator.

The condenser is classified as a sub-cooling condenser and consists of a fin and tube heat exchanger installed between two end tanks. Divisions in the end tanks separate the heat exchanger into a three pass upper (condenser) section and a single pass lower (sub-cooler) section, which are interconnected by the receiver drier module. The desiccant pack and the filter in the receiver drier module are serviceable items retained in position by a threaded plug.

Condenser Assembly



1 Cap
2 Spring clip
3 Sealing plug
4 Desiccant module
5 Condenser
6 Outlet pipe
7 Inlet pipe

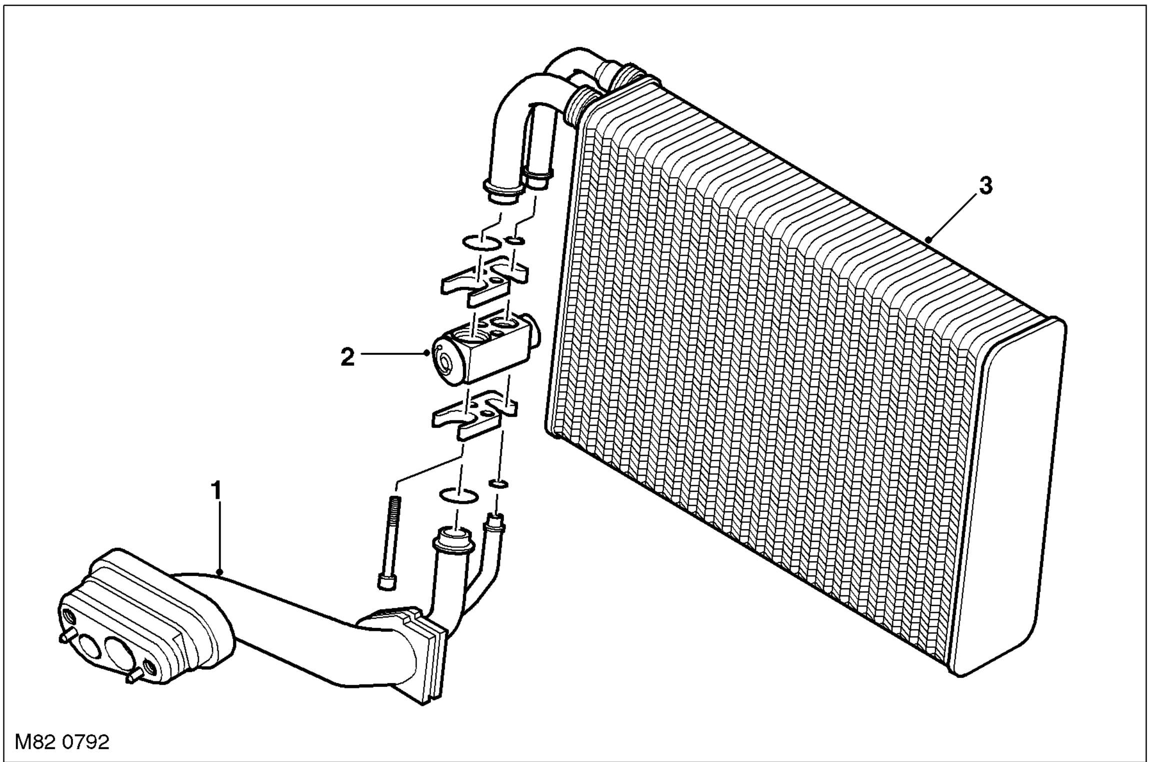
Thermostatic Expansion Valve
The thermostatic expansion valve meters the flow of refrigerant into the evaporator, to match the refrigerant flow with the heat load of the air passing through the evaporator matrix.

The temperature and pressure of the refrigerant leaving the evaporator act on the thermostatic expansion valve to control the volume of refrigerant flowing through the evaporator. The warmer the air flowing through the evaporator matrix, the more heat available to evaporate refrigerant and thus the greater the volume of refrigerant allowed through the metering valve.

Evaporator
The evaporator is installed in the heater assembly between the blower and the heater matrix, to absorb heat from the exterior or recirculated air. Low pressure, low temperature refrigerant changes from liquid to vapour in the evaporator, absorbing large quantities of heat as it changes state.

Most of the moisture in the air passing through the evaporator condenses into water, which drains through the floorpan to the underside of the vehicle through two drain tubes.

Evaporator and Thermostatic Expansion Valve



1 Insulated connecting pipes
2 Thermostatic expansion valve
3 Evaporator

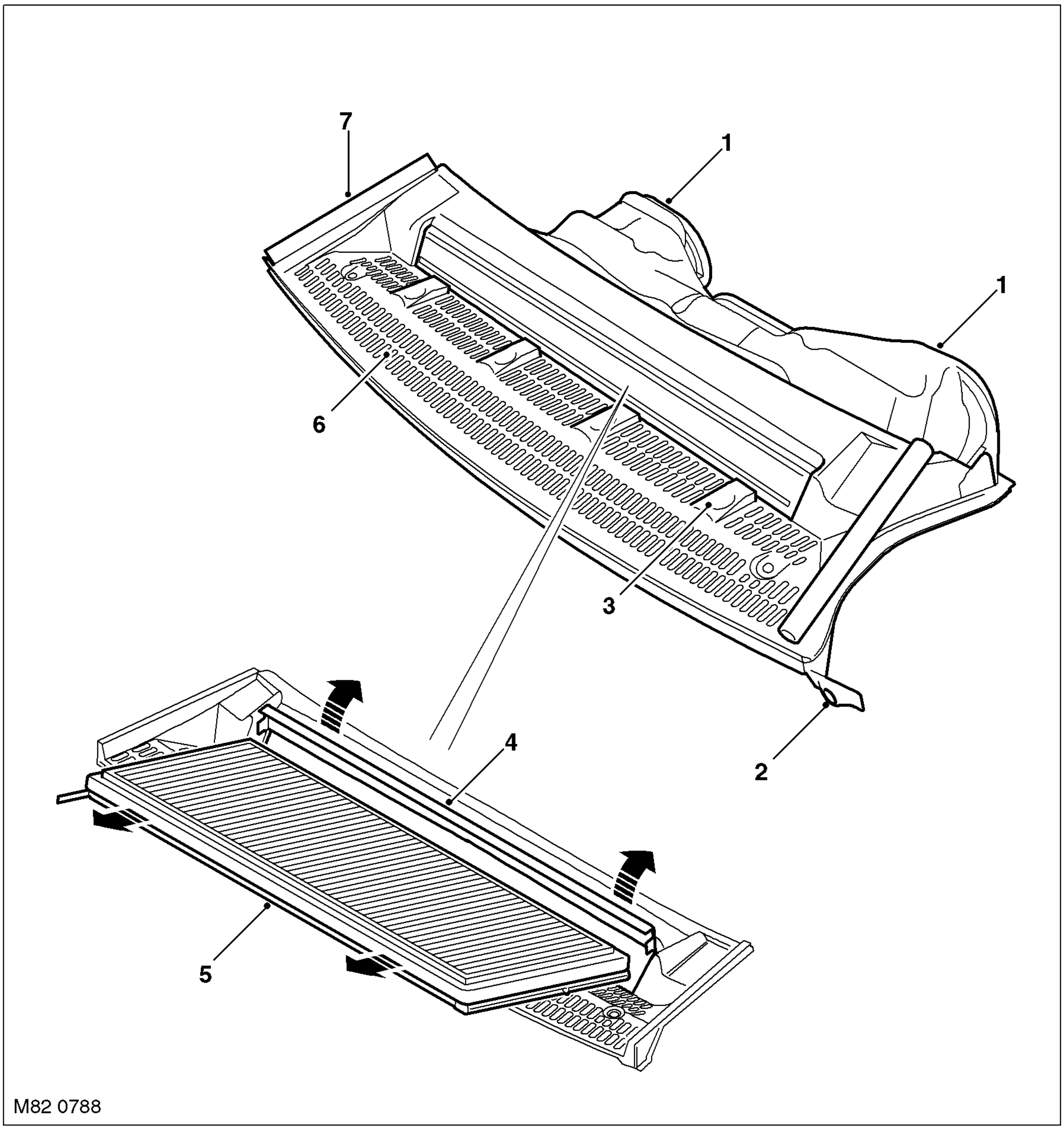
Refrigerant Lines
To maintain similar flow velocities around the system, the diameter of the refrigerant lines varies to suit the two pressure/temperature regimes. The larger diameters are installed in the low pressure/temperature regime and the smaller diameters are installed in the high pressure/temperature regime.

Low and high pressure charging connections are incorporated into the refrigerant lines near the front RH corner of the engine compartment.

Air Inlet Housing
The air inlet housing directs fresh air into the heater unit. The air inlet housing is centrally mounted on the engine bulkhead, below a ventilation grille in the bonnet, and secured to the bulkhead closing panels.

A serviceable particle filter, or particle/odour filter, is installed in the air inlet housing to prevent odours and/or particulate matter from entering the vehicle with the fresh air.

Air inlet Housing



1 Air outlet to heater assembly
2 Fixing lug
3 Filter flap catch
4 Filter flap
5 Particle/carbon filter
6 Inlet grille
7 Seal

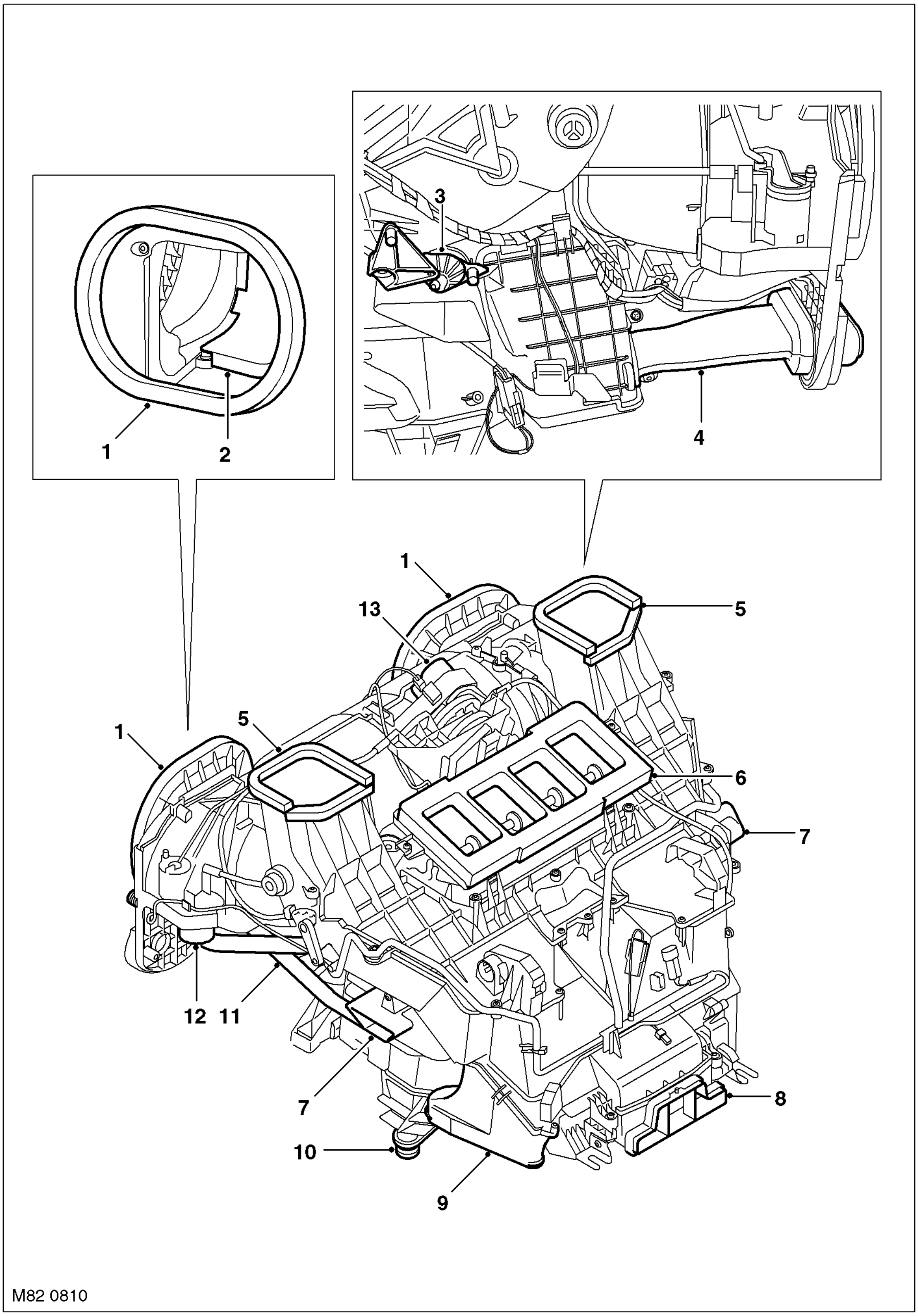
Heater Assembly

The heater assembly controls the temperature, volume and distribution of air supplied to the distribution ducts as directed by selections made on the ATC ECU control panel. The assembly is installed on the vehicle centre-line, between the fascia and the engine bulkhead. The heater assembly consists of a casing formed from a series of plastic mouldings. Internal passages integrated into the casing guide the air through the casing and separate it into two flows, one for the LH outlets and one for the RH outlets. Two drain outlets at the bottom of the casing are connected to overboard drain hoses in the sides of the transmission tunnel.

The heater assembly incorporates:

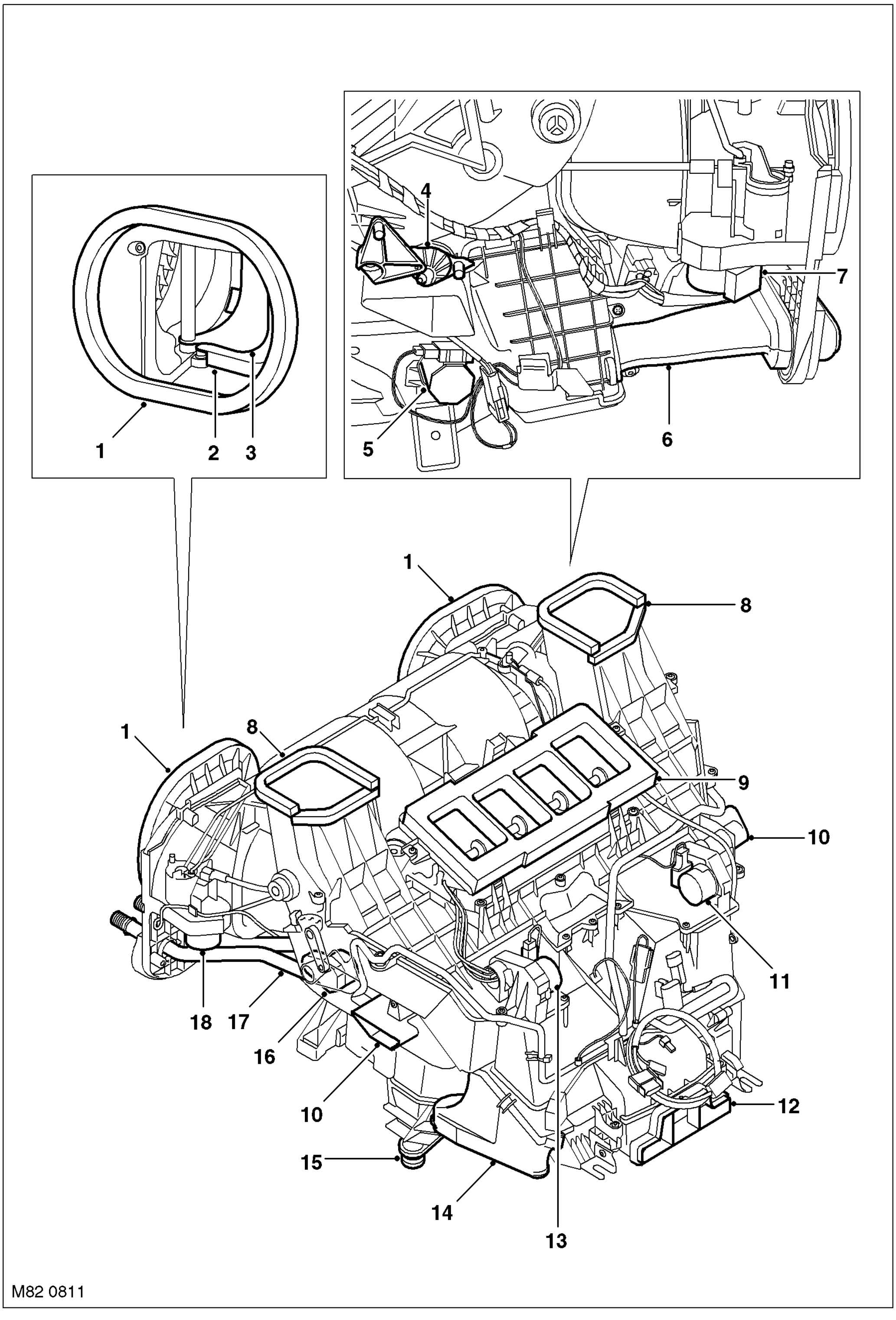
- A blower
- A heater matrix
- Control flaps
- Control flap motors
- The thermostatic expansion valve and the evaporator of the refrigerant system
- The evaporator temperature sensor and either one (low line system) or two (high line system) heater matrix temperature sensors.

Low Line Heater Assembly



1 Fresh air inlet
2 Fresh/Recirculated air flap
3 Face level temperature blend flap control
4 Insulated refrigerant pipes
5 Windscreen air outlet
6 Face level air outlets
7 Front footwell air outlet
8 Rear face level air outlet
9 Rear footwell air outlet
10 Water drain
11 Coolant pipes
12 Fresh/Recirculated air flaps motor
13 Distribution flaps motor

High Line Heater Assembly



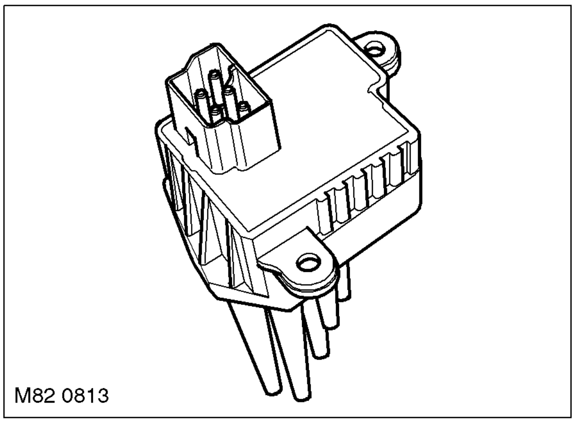
1 Fresh air inlet
2 Fresh/Recirculated air flap
3 Ram air flap
4 Face level temperature blend flap control
5 Rear face level temperature blend motor
6 Insulated refrigerant pipes
7 Ram air flaps motor
8 Windscreen air outlet
9 Face level air outlets
10 Front footwell air outlet
11 Footwell (RH) and face level (LH) air flaps motors
12 Rear face level air outlet
13 Rear footwell air outlet
14 Water drain
15 Windscreen distribution motor
16 Coolant pipes
17 Fresh/Recirculated air flap motor

Blower

The blower is installed between the air inlets and the evaporator, and consists of two open hub, centrifugal fans powered by a single electric motor. Operation of the electric motor is controlled by the ATC ECU via an output stage (voltage amplifier) installed in the outlet of the RH fan.

To produce the seven blower speeds the ATC ECU outputs a stepped control voltage between 0 and 8 V to the output stage, which regulates a battery power feed from the passenger compartment fusebox to the blower. The control voltage changes, in 1 V steps, between 2 V (blower speed 1) and 8 V (blower speed 7). If the control voltage is less than 2 V the blower is off.

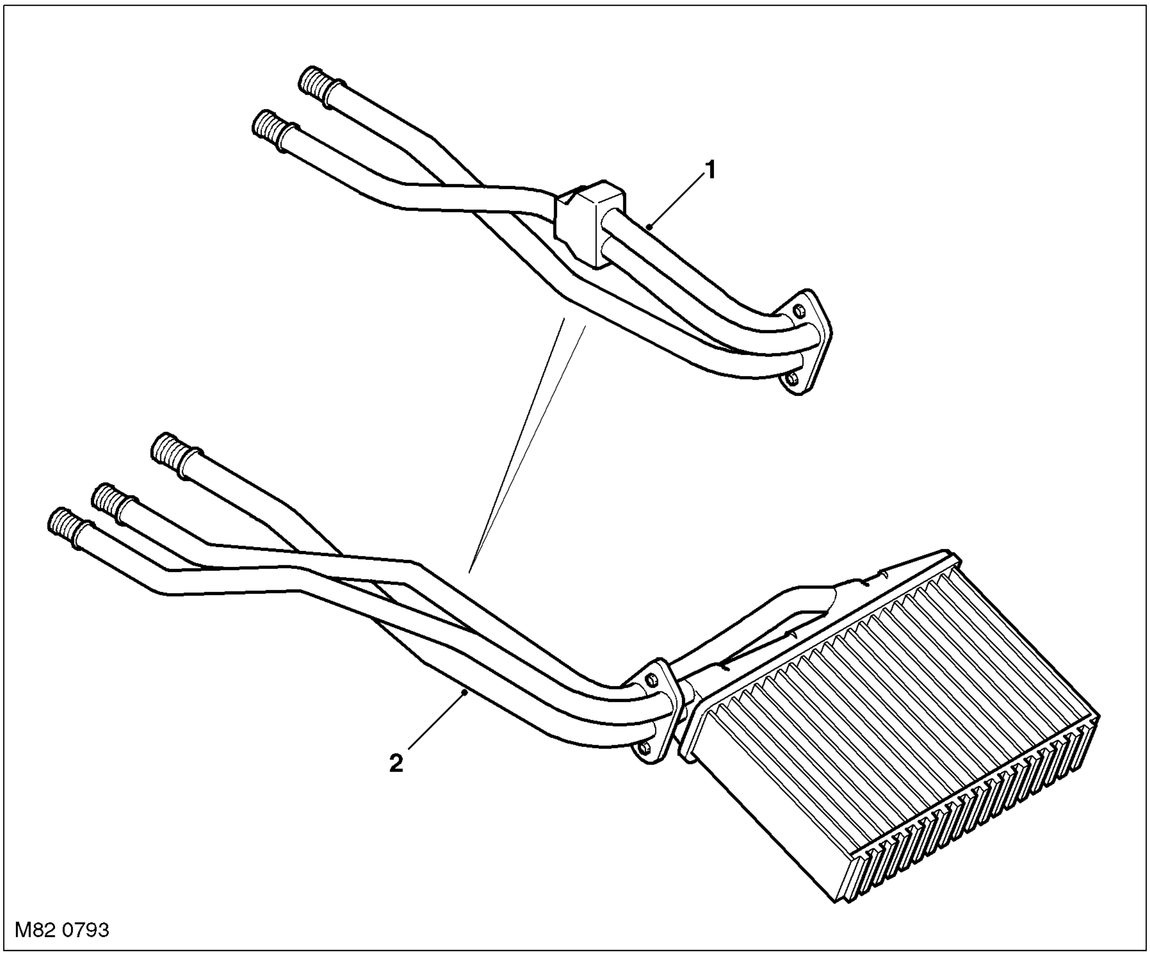
Blower Output Stage



Heater matrix

The same heater matrix is used in both the low and high line heater assemblies. The heater matrix is internally divided into two separate halves, with separate coolant inlets for each half and a common coolant outlet. On the low line system, the two coolant inlets are connected to a common feed from the single coolant valve. On the high line system, each coolant inlet pipe is connected to a feed from a separate coolant valve.

Heater Matrix Assembly



1 Low line connecting pipes

2 High line connecting pipes

Control Flaps

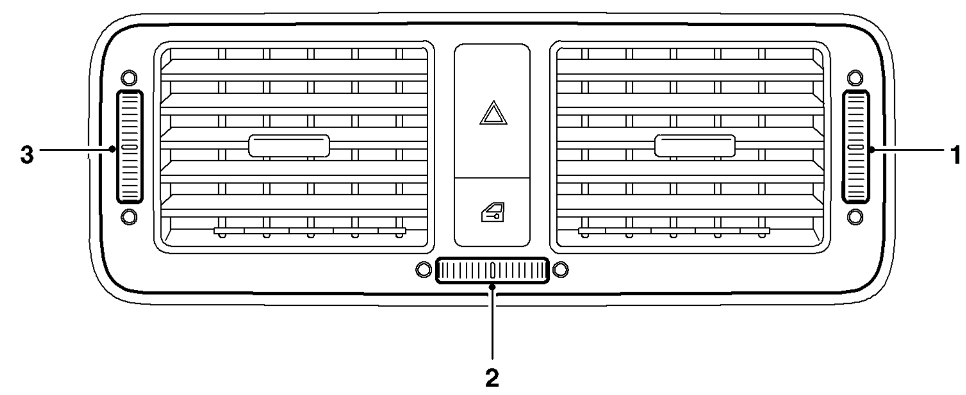
Control flaps in the heater assembly control the source of inlet air and the distribution and temperature of outlet air.

On both the low and high line heater assemblies, a fresh/recirculated air flap is installed in the air inlet on each side of the heater assembly. A stepper motor drives the LH fresh/recirculated air flap and a Bowden cable transmits the drive from the LH to the RH fresh/recirculated air flap. On the high line system, a ram air flap is installed inside each fresh/recirculated air flap. A stepper motor drives the RH ram air flap and a Bowden cable transmits the drive from the RH to the LH ram air flap.

Each side of the heater assembly contains separate distribution flaps for the footwell, face level and windscreen. The related flaps on each side of the heater assembly are installed on common drive spindles. On the low line heater assembly, the distribution flaps are driven by Bowden cables connected to a cam mechanism, which, in turn, is driven by a stepper motor. On the high line heater assembly, each set of distribution flaps is driven by a separate stepper motor.

On both the low and high line heater assemblies, a blend flap is installed below the face level outlets. The blend flap is driven by a Bowden cable connected to a thumbwheel on the centre face vents in the fascia, and allows the temperature of face level air to be modified with cold air direct from the evaporator.

Fascia Face Level Vent

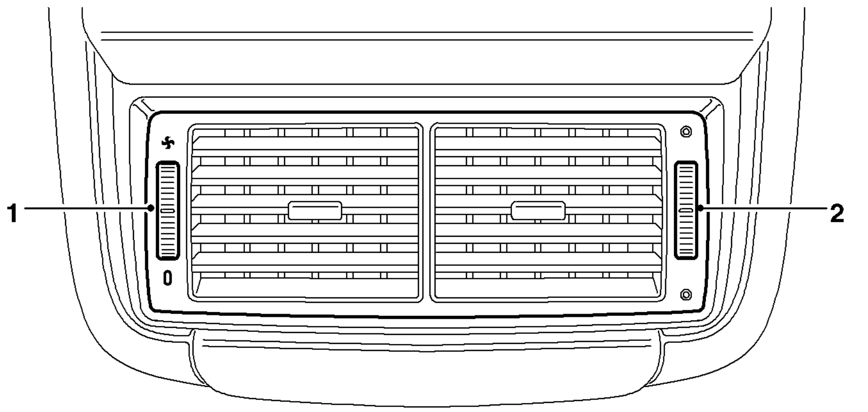


M82 0817

1 RH air control thumbwheel
2 Temperature control thumbwheel
3 LH air control thumbwheel

On high line heater assemblies, a blend flap is installed for the rear passenger face level outlet to allow the temperature of rear face level air to be adjusted independently from the temperatures selected on the control panel. The blend flap is driven by a stepper motor controlled by a thumbwheel on the rear passenger face vent. The blend flap is also used to close off the rear face level outlet when maximum air output is required for the front outlets, e.g. when defrost is selected.

Rear Passenger Face Level Vent

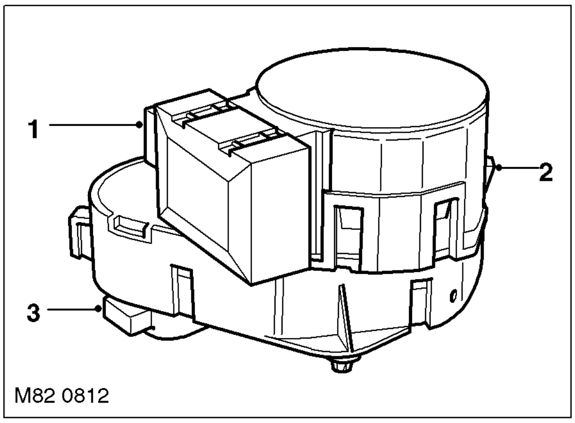


M82 0818

1 Blower control
2 Temperature control

Control Flap Motors
Two types of electrical stepper motor are used to operate the control flaps in the heater assembly. A conventional 500 Hz stepper motor operates the recirculation flaps, on the low and high line systems. On the high line system, five bus controlled 200 Hz stepper motors operate the ram air, distribution (windscreen, face level and footwell) and the rear face level temperature control flaps. On the low line system a bus controlled stepper motor operates the distribution flaps cam mechanism. All of the stepper motors are controlled by the ATC ECU. None of the stepper motors are interchangeable.

Typical Control Flap Motor



1 Electrical connector
2 Release clip
3 Output shaft

Each bus controlled stepper motor incorporates a microprocessor and is connected to an M bus from the ATC ECU, which consists of three wires making up power, ground and signal circuits. The microprocessor in each bus controlled stepper motor is programmed with a different address. Each M bus message from the ATC ECU contains the address of an individual bus controlled stepper motor, so only that motor responds to the message.

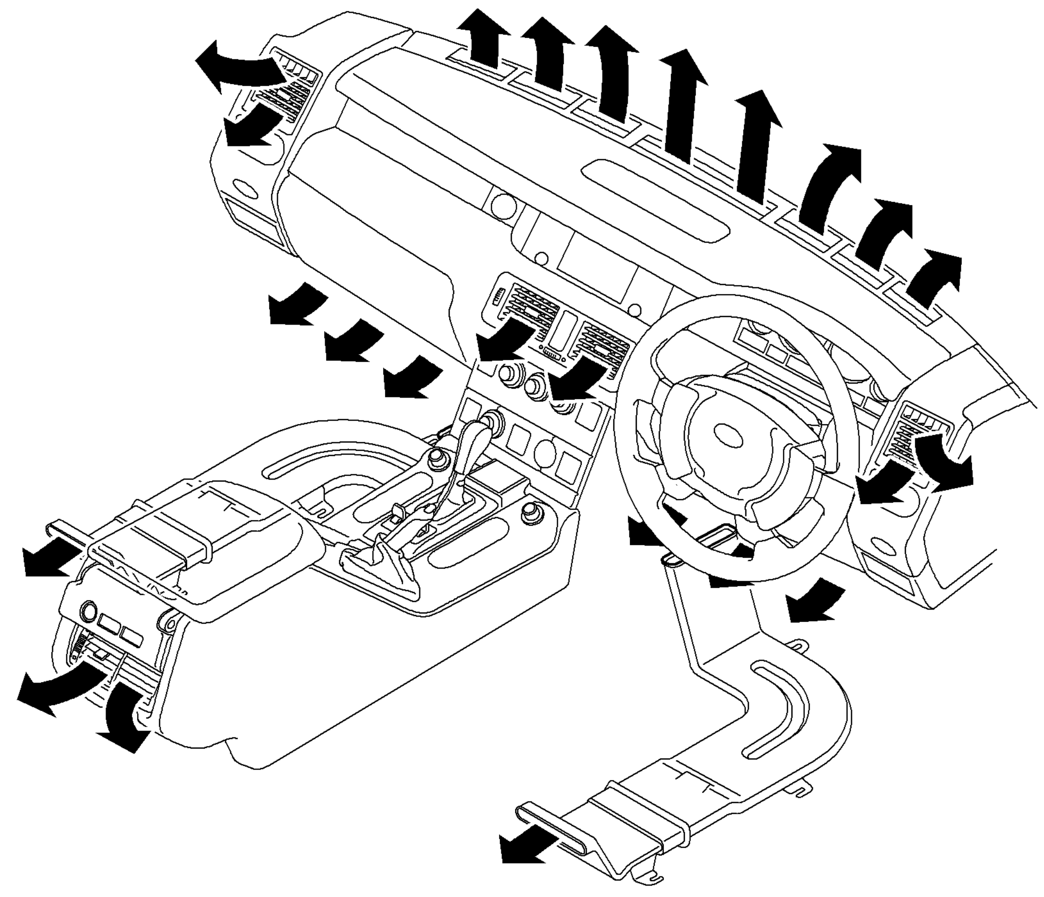
None of the stepper motors incorporate a feedback potentiometer. Instead, the ATC ECU determines the positions of the flaps by using either their closed or open position as a datum and memorising the steps that it drives the individual stepper motors. Each time the ignition is switched on, the ATC ECU checks the memorised position of the stepper motors against fixed values for the current system configuration. If there is an error (e.g. after a power supply failure during operation or after replacement of the ATC ECU), the ATC ECU calibrates the applicable stepper motors, to re-establish the datums, by driving them fully closed or open before re-setting them to their nominal position. A calibration run can also be invoked using TestBook/T4.

When any of the control flaps are set to fully closed or open, the ATC ECU signals the related stepper motor to move the appropriate number of steps in the applicable direction. To accommodate build tolerances and wear, and to ensure the flaps are held in the selected position, every 20 seconds the ATC ECU signals the stepper motor to move an additional 10 steps in the relevant direction.

Distribution Ducts
Air from the heater assembly is distributed around the vehicle interior through distribution ducts to outlets in the fascia, the front and rear footwells, and the rear of the cubby box between the front seats.

In the fascia, the distribution ducts are connected to fixed vents for the windscreen and side windows and adjustable vent assemblies for face level air. An adjustable vent assembly is also installed on the rear of the cubby box for rear face level air. The footwell outlets are fixed vents formed in the end of the related distribution ducts.

Air Distribution



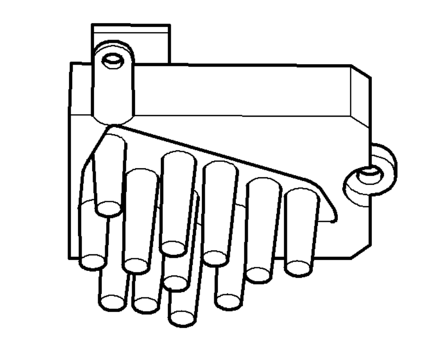
Forced Ventilation Outlets
The forced ventilation outlets promote the free flow of air through the passenger compartment. The outlets are installed in the LH and RH rear quarter body panels and vent passenger compartment air into the sheltered area between the rear quarter body panels and the rear bumper.

Rear Blower (High Line System Only)
The blower is installed between the front seats, in the rear face air duct, and consists of an open hub, centrifugal fan powered by an electric motor. Operation of the electric motor is controlled by a thumbwheel on the rear passenger face vent via the ATC ECU and an output stage (voltage amplifier) installed in the outlet of the fan.

The thumbwheel operates a variable potentiometer which outputs between 1.25 V (blower off) and 5 V (maximum blower speed) to the ATC ECU. The ATC ECU then outputs a proportional control voltage between 0 and 5 V to the output stage, which regulates a battery power feed from the rear blower relay to the blower to produce the related blower speed.

The rear blower relay is installed in the rear fusebox and energised while the ignition is on.

Rear Blower Output Stage



M82 0814

ATC ECU

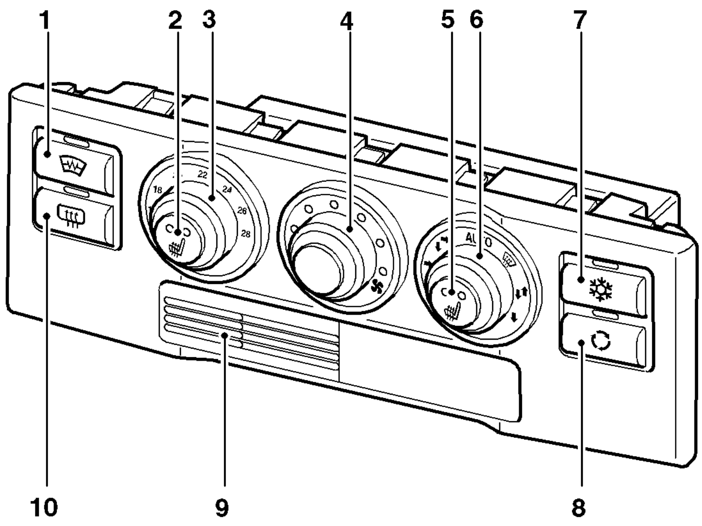
The ATC ECU is installed in the centre of the fascia. An integral control panel contains push switches and rotary switches/knobs for system control inputs. Orange tell-tale LED's in the switches and switch surrounds illuminate to indicate the current settings of the system. The rotary temperature switch is graduated in degrees Celsius, except on USA vehicles, where it is graduated in degrees Fahrenheit.

An in-car temperature sensor and associated electric fan are installed behind a grille in the control panel.

The ATC ECU processes inputs from the control panel switches and system sensors, then outputs the appropriate signals to control the A/C system. In addition to controlling the A/C system, the ATC ECU also controls the following:

- The windscreen heater and windscreen wiper parking area heater (optional fit, not available on vehicles with infra red protection glass)
- The windscreen washer jet heaters
- The rear window heater
- The front seat heaters.

Low Line Control Panel



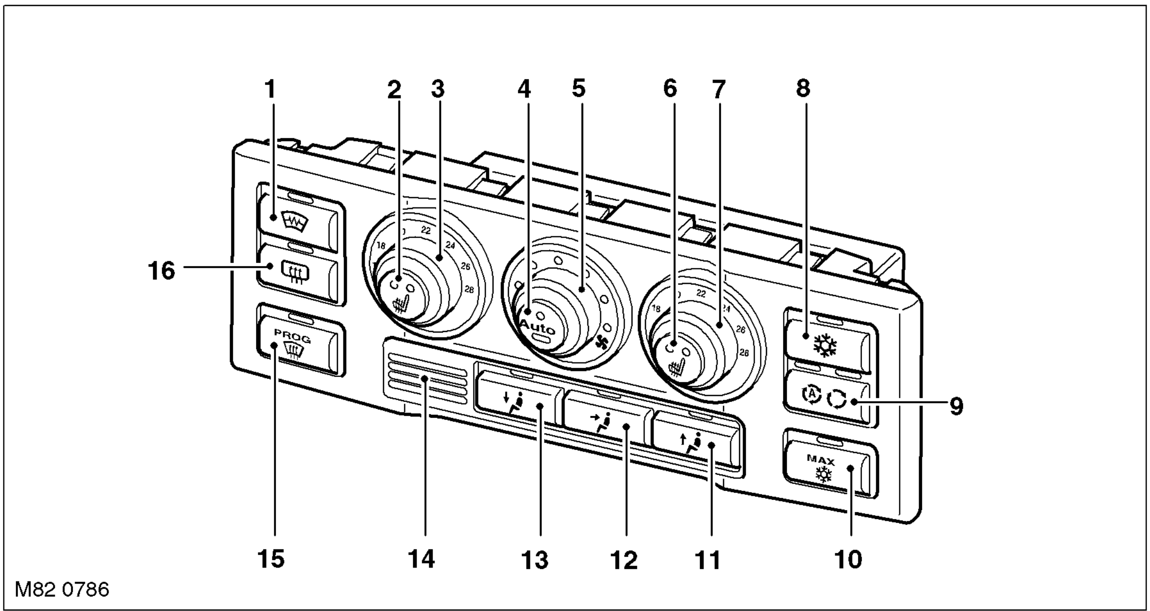
M82 0785

1 Windscreen heater switch
2 LH front seat heater switch
3 Temperature switch
4 Blower switch
5 RH front seat heater switch
6 Distribution control knob
7 A/C switch
8 Recirculation switch
9 In-car temperature sensor
10 Rear window heater switch

The switches on the low line control panel have the following functions:

- *Windscreen Heater Switch:*Enabled only with the engine running. Pressing the switch energises the windscreen heater and the windscreen wiper parking area heater for a set time period, until the switch is pressed again or until the engine stops, whichever occurs first. A LED above the switch is illuminated while the heaters are on.
- *LH and RH Seat Heater Switches:*Activates the heater elements in the seat cushion and seat back at one of two heat levels. The first press of the switch energises the heater elements at the higher heat setting and illuminates two LED in the switch. A second press of the switch sets the heater elements to the lower heat setting and extinguishes one of the LED's. A further press of the switch de-energises the heater elements and extinguishes the second LED. The seat heaters remain on until selected off or the ignition is switched off.
- *Temperature Switch:* Adjusts the passenger compartment nominal temperature setting between 16 and 28 °C (60 and 84 °F). The temperature range is engraved on the switch surround (°F for NAS, °C for rest of world). A pointer on the switch indicates the selected temperature. In the minimum and maximum temperature positions, the temperature is set to maximum cooling or maximum heating respectively.
- *Blower Switch:* For manual adjustment of blower speed. Up to seven LED's in the switch surround illuminate to indicate the selected blower speed.
- *Distribution Control Knob:* For manual adjustment of air distribution. Includes AUTO setting where distribution flaps are set to a nominal comfort position.
- *A/C Switch:* Activates the A/C compressor. Allows the A/C compressor to be selected off for economy operation. A LED above the switch is illuminated when the A/C compressor is selected on.
- *Recirculation Switch:* For manual selection of fresh or recirculated air. A LED above the switch is illuminated when recirculated air is selected.
- *Rear Window Heater Switch:* Enabled only with the engine running. Pressing the switch energises the rear window heater for a set time period, until the switch is pressed again or until the engine stops, whichever occurs first. A LED above the switch is illuminated while the heater is on.

High Line Control Panel



1 Windscreen heater switch
2 LH front seat heater switch
3 LH temperature switch
4 Automatic mode switch
5 Blower switch
6 RH front seat heater switch
7 RH temperature switch
8 A/C switch
9 Recirculation switch
10 Maximum A/C switch
11 Windscreen distribution switch
12 Face distribution switch
13 Footwell distribution switch
14 In-car temperature sensor
15 Defrost programme switch
16 Rear window heater switch

The switches on the high line control panel have the following functions:

- *Windscreen Heater Switch:*Energises the windscreen heater and the windscreen wiper parking area heater for a set time period, until the switch is pressed again or until the engine stops, whichever occurs first. A LED above the switch is illuminated while the heaters are on.
- *LH and RH Seat Heater Switches:*Activates the heater elements in the seat cushion and seat back at one of two heat levels. The first press of the switch energises the heater elements at the higher heat setting and illuminates two LED in the switch. A second press of the switch sets the heater elements to the lower heat setting and extinguishes one of the LED's. A further press of the switch de-energises the heater elements and extinguishes the second LED. The seat heaters remain on until selected off or the ignition is switched off.
- *LH and RH Temperature Switches:*Adjusts the nominal temperature settings of the LH and RH sides of the passenger compartment between 16 and 28 °C (60 and 84 °F). The temperature range is engraved on the switch surrounds. A pointer on each switch indicates the selected temperature. In the minimum and maximum temperature positions, the system operates at maximum cooling or maximum heating respectively.
- *Automatic Mode Switch:* Activates the automatic modes for air volume and distribution and also activates the compressor. Separate LED's in the automatic mode switch illuminate when the blower and the distribution control flaps are in automatic mode. Manually selecting the blower speed or a distribution switch extinguishes the related LED.
- *Blower Switch:* For manual adjustment of blower speed. Up to seven LED's in the switch surround illuminate to indicate the selected blower speed.
- *A/C Switch:* Controls activation of the A/C compressor. Allows the A/C compressor to be selected off for economy operation. A LED above the switch is illuminated when the A/C compressor is selected on.
- *Recirculation Switch:* For manual or automatic selection of fresh or recirculated air. Two LED's above the switch illuminate to indicate the mode and position of the recirculation flaps. The first press of the switch sets the recirculation flaps to automatic mode and illuminates the LH LED. A second press of the switch manually sets the recirculation flaps to the recirculation position, extinguishes the LH LED and illuminates the RH LED. A further press of the switch manually sets the recirculation flaps to the fresh air position and extinguishes the RH LED.

- *Maximum A/C Switch:* For selection of maximum A/C when the ignition is on or rest heating when the ignition is off. A LED above the switch is illuminated when maximum cooling or rest heating is selected.
- *Distribution Switches (Windscreen, Face and Footwell):* For manual selection of air distribution in any combination of windscreen, face and footwell outlets. A LED above each switch illuminates when a selection is made.
- *Defrost Programme Switch:* Activates a programme that automatically selects the windscreen heater on, activates the compressor and changes the system settings to direct dry heat to the windscreen. A LED above the switch is illuminated while the defrost programme is active.
- *Rear Window Heater Switch:* Enabled only with the engine running. Pressing the switch energises the rear window heater for a set time period, until the switch is pressed again or until the engine stops, whichever occurs first. A LED above the switch is illuminated while the heater is on.

Inputs and Outputs

Five electrical connectors provide the interface between the ATC ECU and the vehicle/heater assembly wiring.

Both the low and high line systems receive ambient temperature, engine coolant temperature, engine speed and vehicle speed inputs in K bus messages from the instrument pack. If the K bus messages are missing or faulty, the ATC ECU adopts the following default values:

- Ambient temperature = 0 °C (32 °F)
- Engine coolant temperature = 80 °C (176 °F)
- Engine speed = 800 rev/min
- Vehicle speed = zero.

If a fault develops in the input from the temperature selector switch on the control panel, the ATC ECU adopts a default value of 24 °C (75 °F).

ATC ECU Harness Connector C0249 Pin Details

Pin No.	Description	System		Input/Output
		Low	High	
1	Front seat heating battery power supply	Yes	Yes	Input
2	LH front seat heater	Yes	Yes	Output
3	RH front seat heater	Yes	Yes	Output

ATC ECU Harness Connector C0923 Pin Details

Pin No.	Description	System		Input/Output
		Low	High	
1	Rear blower switch signal	No	Yes	Input
2	Rear blower switch power supply	No	Yes	Output
3	Automatic distribution switch	Yes	No	Input
4	LH heater matrix temperature sensor	No	Yes	Input
5	RH heater matrix temperature sensor	Yes	Yes	Input
6	Evaporator temperature sensor	Yes	Yes	Input
7	Blower control voltage	Yes	Yes	Output
8	Rear temperature switch	No	Yes	Input
9	Sensor ground (evaporator, and heater temperature sensors)	Yes	Yes	–
10 to 12	Not used	–	–	–
13	Recirculation flap motor signal 2	Yes	Yes	Input
14	Recirculation flap motor signal 2	Yes	Yes	Output
15	Recirculation flap motor signal 1	Yes	Yes	Output
16	Recirculation flap motor signal 1	Yes	Yes	Input
17	Rear blower and temperature switch power supply	No	Yes	Output
18	Not used	–	–	–

ATC ECU Harness Connector C1629 Pin Details

Pin No.	Description	System		Input/Output
		Low	High	
1	Ignition power supply	Yes	Yes	Input
2	Sunlight sensor ground	No	Yes	–
3	K bus	Yes	Yes	Input/Output
4	Delayed accessory power	Yes	Yes	Input
5	LH coolant valve	Yes	Yes	Output
6	RH coolant valve	No	Yes	Output
7	HRW relay	Yes	Yes	Output
8	Windscreen washer jet heater relay	Yes	Yes	Output
9	Auxiliary coolant pump	Yes	Yes	Output
10	Refrigerant pressure sensor signal	Yes	Yes	Input
11	Not used	–	–	–
12	Pollution sensor signal	No	Yes	Input
13	Pollution sensor power supply	No	Yes	Output
14	Windscreen heater relay	Yes	Yes	Output
15	LH seat heating temperature sensor	Yes	Yes	Output
16	RH seat heating temperature sensor	Yes	Yes	Output
17	Engine full load signal	Yes	Yes	Input
18	LH sunlight sensor signal	No	Yes	Input
19	RH sunlight sensor signal	No	Yes	Input
20	Refrigerant pressure sensor ground	Yes	Yes	–
21	Refrigerant pressure sensor power supply	Yes	Yes	Output
22	Coolant changeover valve	Yes	Yes	Output
23	Instrument illumination	Yes	Yes	Input
24	Sunlight sensor power supply	No	Yes	Output
25	Pollution sensor heater ground	No	Yes	–
26	Pollution sensor heater power supply	No	Yes	Output

ATC ECU Harness Connector C1630 Pin Details

Pin No.	Description	System		Input/Output
		Low	High	
1	Battery power supply	Yes	Yes	Input
2	A/C compressor clutch	Yes	Yes	Output
3	System ground	Yes	Yes	–

ATC ECU Harness Connector C2295 Pin Details

Pin No.	Description	System		Input/Output
		Low	High	
1	M bus power supply	No	Yes	Output
2	M bus ground	No	Yes	–
3	M bus interface line	No	Yes	Input/Output

Sensors

Both the low and the high line systems incorporate:

- An in-car temperature sensor
- A refrigerant pressure sensor
- An evaporator temperature sensor
- A heater matrix temperature sensor.

The high line system also incorporates:

- A second heater matrix temperature sensor
- A sunlight sensor
- A pollution sensor.

In-car Temperature Sensor

The in-car temperature sensor is an encapsulated Negative Temperature Coefficient (NTC) thermistor that provides the ATC ECU with an input of passenger compartment air temperature. The in-car temperature sensor is installed behind a grille in the ATC ECU control panel. An electric fan in the ATC ECU runs continuously, while the ignition is on, to draw air through the grille and across the in-car temperature sensor.

The ATC ECU uses the signal from the in-car temperature sensor for control of the coolant temperature valve(s), blower speed and air distribution.

The signal voltage from the in-car temperature sensor is between 0 and 5 V. The ATC ECU monitors the signal voltage and defaults to a temperature of 20 °C (68 °F) if it goes out of the range 0.573 – 4.882 V:

- If the signal voltage is less than 0.573 V, the ATC ECU assumes there is a short circuit to ground
- If the signal voltage is more than 4.882 V, the ATC ECU assumes there is an open circuit or a short circuit to battery.

Refrigerant Pressure Sensor

The refrigerant pressure sensor provides the ATC ECU with a pressure input from the high pressure side of the refrigerant system. The refrigerant pressure sensor is located in the refrigerant line between the condenser and the thermostatic expansion valve.

The ATC ECU supplies a 5 V reference voltage to the refrigerant pressure sensor and receives a return signal voltage, between 0 and 5 V, related to system pressure.

The ATC ECU uses the signal from the refrigerant pressure sensor to protect the system from extremes of pressure and to calculate compressor load on the engine for idle speed control. The ATC ECU also transfers the pressure value to the Engine Control Module (ECM), via the K bus, instrument pack and CAN bus, for use in controlling the speed of the engine cooling fan.

To protect the system from extremes of pressure, the ATC ECU disengages the compressor clutch if the pressure:

- Decreases to 1.9 ± 0.2 bar (27.5 ± 3 lbf/in^2): the ATC ECU engages the compressor clutch again when pressure increases to 2.8 ± 0.2 bar (40.5 ± 3 lbf/in^2)
- Increases to 33 ± 1 bar (479 ± 14.5 lbf/in^2): the ATC ECU engages the compressor clutch again when pressure decreases to 23.5 ± 1 bar (341 ± 14.5 lbf/in^2).

Evaporator Temperature Sensor

The evaporator temperature sensor is a NTC thermistor that provides the ATC ECU with a temperature signal from the air outlet side of the evaporator. The evaporator temperature sensor is installed in the RH side of the heater assembly casing, and extends into the core of the evaporator.

The ATC ECU uses the input from the evaporator temperature sensor to control the engagement and disengagement of the compressor clutch, to prevent the formation of ice on the evaporator.

The signal voltage from the evaporator temperature sensor is between 0 and 5 V. The ATC ECU monitors the signal voltage and defaults to a temperature of 0 °C (32 °F) if it goes out of the range 0.157 – 4.784 V:

- If the signal voltage is less than 0.157 V, the ATC ECU assumes there is a short circuit to ground
- If the signal voltage is more than 4.784 V, the ATC ECU assumes there is an open circuit or a short circuit to battery.

Heater Matrix Temperature Sensor

The heater matrix temperature sensor is a NTC thermistor that provides the ATC ECU with a temperature signal from the air outlet side of the heater matrix. On low line systems, a single sensor is installed in the rear of the heater assembly casing, to the right of the centre-line. On high line systems, two sensors are installed, one each side of the centre-line.

The ATC ECU uses the input from the heater matrix temperature sensor(s) to control the operation of the coolant valve(s).

The signal voltage from each heater matrix temperature sensor is between 0 and 5 V. The ATC ECU monitors the signal voltage and defaults to a temperature of 55 °C (131 °F) if it goes out of the range 0.173 – 4.890 V:

- If the signal voltage is less than 0.173 V, the ATC ECU assumes there is a short circuit to ground
- If the signal voltage is more than 4.890 V, the ATC ECU assumes there is an open circuit or a short circuit to battery.

Sunlight Sensor

The sunlight sensor consists of two photoelectric cells that provide the ATC ECU with inputs of light intensity, one as sensed coming from the left of the vehicle and one as sensed coming from the right. The inputs are a measure of the solar heating effect on vehicle occupants and used by the ATC ECU to adjust blower speed, temperature and distribution to improve comfort. The sensor is installed in the centre of the fascia upper surface.

If one of the photoelectric cells is faulty, the output from the other photoelectric cell is used for both sides of the vehicle. If both photoelectric cells are faulty, the ATC ECU uses a default value of zero.

Pollution Sensor

The pollution sensor allows the ATC ECU to monitor the ambient air for the level of hydrocarbons and oxidized gases such as nitrous oxides, sulphur oxides and carbon monoxide. The pollution sensor is installed at the rear of the radiator, on the upper RH side of the viscous fan housing.

The ATC ECU outputs a battery power supply to heat the pollution sensor to operating temperature, and a 5 V reference voltage for the signal. The signal voltage from the pollution sensor is between 0 and 5 V.

If there is a fault with the pollution sensor, the ATC ECU disables automatic closing of the recirculation flaps on detection of pollutants.

Auxiliary Coolant Pump

The auxiliary coolant pump is an electric pump that ensures there is a satisfactory flow rate through the heater matrix at low engine speeds. The auxiliary coolant pump is installed in the engine compartment, in a rubber mounting attached to the side of the LH suspension turret. Operation of the auxiliary coolant pump is controlled by a power supply from the ATC ECU.

Coolant Valve

The coolant valve controls the coolant flow to the heater matrix. A single coolant valve controls the coolant flow to both sides of the heater matrix on low line systems. On high line systems, separate coolant valves control the coolant flow to each side of the heater matrix. The coolant valves are installed in the engine compartment on a bracket attached to the side of the LH suspension turret.

Each coolant valve is a normally open solenoid valve controlled by a Pulse Width Modulated (PWM) signal from the ATC ECU. The ATC ECU changes the length of time the coolant valve is open each duty cycle between 0 second (valve closed) and 3.6 seconds (valve held open). On the high line system, the PWM signals to the two valves are phase offset by 1.8 seconds to reduce coolant flow fluctuations.

FBH System

The system consists of a FBH unit, a FBH fuel pump and a changeover valve. On vehicles with the remote operation feature, the system also includes a FBH receiver and a remote handset.

Fuel for the FBH system is taken from the vehicle fuel tank, through a line attached to the fuel tank's fuel pump unit, and supplied via the FBH fuel pump to the FBH unit. In the FBH unit, the fuel delivered by the FBH fuel pump is burned and the resultant heat output is used to heat the engine coolant. The changeover valve isolates the heater coolant circuit from the engine coolant circuit

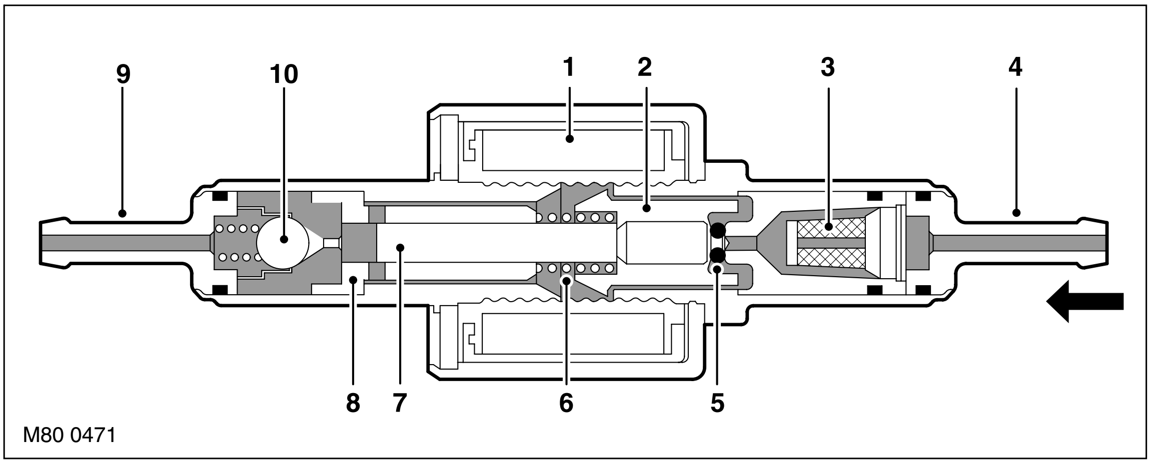
An ECU integrated into the FBH unit controls the operation of the FBH unit and the FBH fuel pump. The ATC ECU controls the changeover valve. System operation is initiated by:

- The instrument pack, via the ATC ECU, using I and K bus messages, for parked heating selections made on the Multi Information Display (MID) or Multi-Function Display (MFD)
- The remote handset, via the TV antenna, TV amplifier and the FBH receiver, using radio and hardwired signals, for instant activation of parked heating
- The ATC ECU, using K bus messages, for additional heating while the engine is running.

FBH Fuel Pump

The FBH fuel pump regulates the fuel supply to the FBH unit. The FBH fuel pump is installed below the RH side of the fuel tank in a rubber mounting attached to the rear subframe. The pump is a self priming, solenoid operated plunger pump. The ECU in the FBH unit outputs a pulse width modulated signal to control the operation of the pump. When the pump is de-energised, it provides a positive shut-off of the fuel supply to the FBH unit.

Sectioned View of FBH Fuel Pump



1 Solenoid coil
2 Plunger
3 Filter insert
4 Fuel line connector
5 'O' ring seal
6 Spring
7 Piston
8 Bush
9 Fuel line connector
10 Non return valve

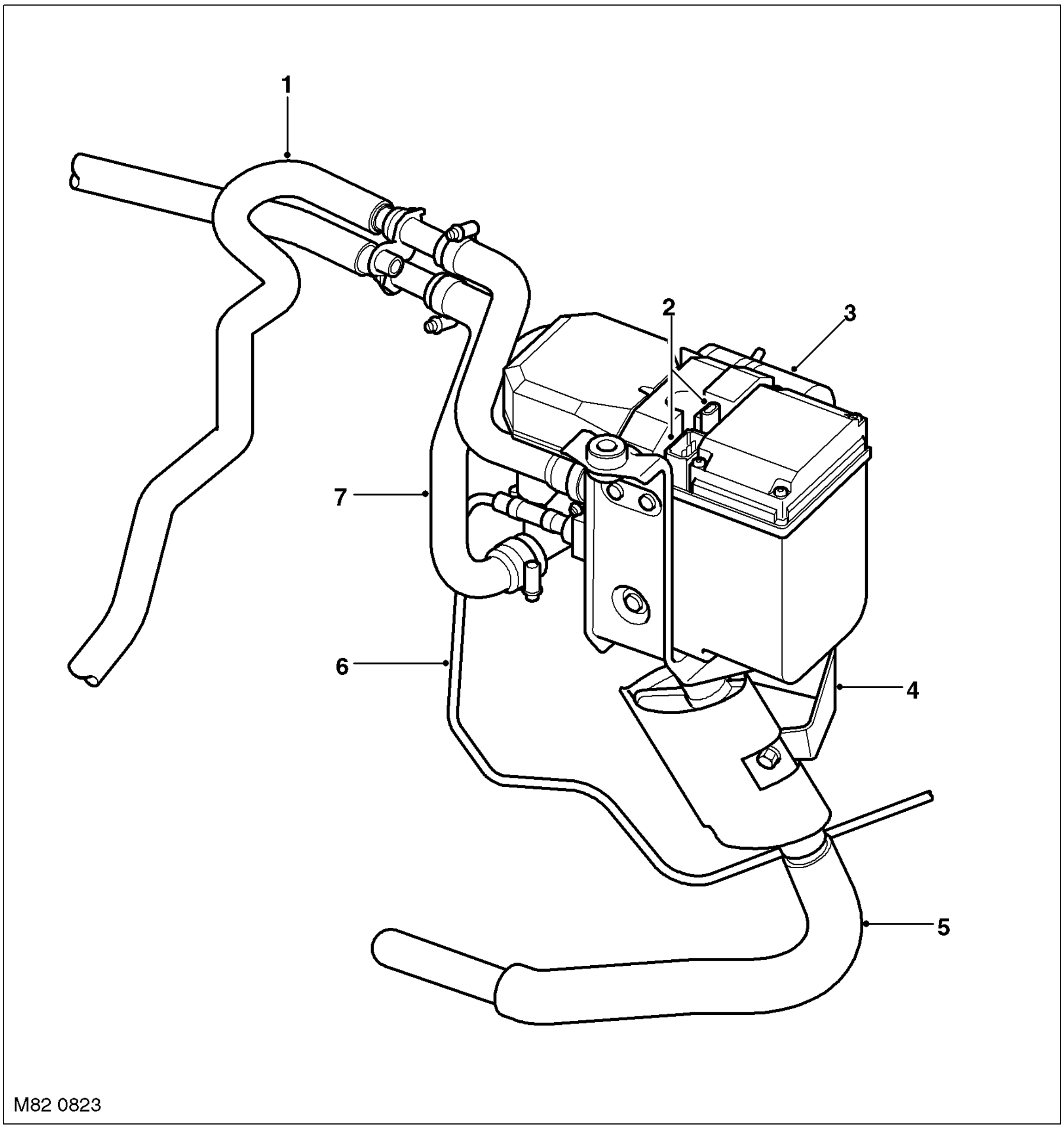
The solenoid coil of the FBH fuel pump is installed around a housing which contains a plunger and piston. The piston locates in a bush, and a spring is installed on the piston between the bush and the plunger. A filter insert and a fuel line connector are installed in the inlet end of the housing. A non return valve and a fuel line connector are installed in the fuel outlet end of the housing.

While the solenoid coil is de-energised, the spring holds the piston and plunger in the 'closed' position at the inlet end of the housing. An 'O' ring seal on the plunger provides a fuel tight seal between the plunger and the filter insert, preventing any flow through the pump. When the solenoid coil is energised, the piston and plunger move towards the outlet end of the housing, until the plunger contacts the bush; fuel is then drawn in through the inlet connection and filter. The initial movement of the piston also closes transverse drillings in the bush and isolates the pumping chamber at the outlet end of the housing. Subsequent movement of the piston then forces fuel from the pumping chamber through the non return valve and into the line to the FBH unit. When the solenoid de-energises, the spring moves the piston and plunger back towards the closed position. As the piston and plunger move towards the closed position, fuel flows past the plunger and through the annular gaps and transverse holes in the bush to replenish the pumping chamber.

FBH Unit

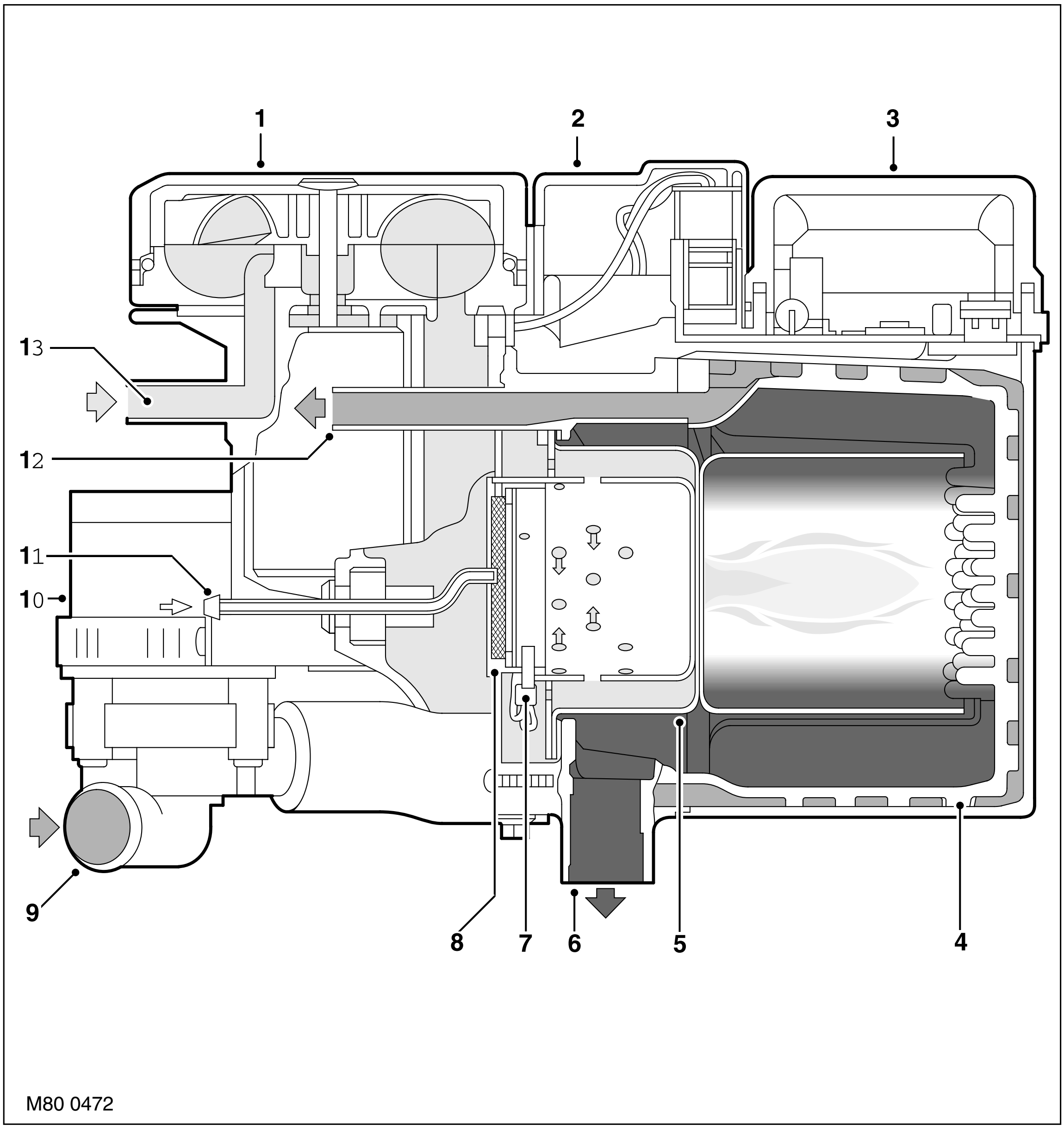
The FBH unit is installed in the passenger side rear of the engine compartment, below the battery. It is connected in series with the coolant supply to the heater assembly. Two electrical connectors on the FBH unit connect it to the vehicle wiring.

FBH Unit Components



1 Coolant outlet hose
2 Electrical connectors
3 Air inlet filter
4 Mounting bracket
5 Exhaust pipe
6 Fuel supply line
7 Coolant inlet hose

Sectioned View of FBH Unit



1	Combustion air fan	**8**	Evaporator
2	Burner housing	**9**	Coolant inlet
3	ECU	**10**	Circulation pump
4	Heat exchanger	**11**	Fuel inlet
5	Burner insert	**12**	Coolant outlet
6	Exhaust	**13**	Air inlet
7	Glow plug/flame sensor		

The FBH unit consists of:

- A circulation pump
- A combustion air fan
- A burner housing
- An ECU/heat exchanger
- An air inlet hose
- An exhaust pipe
- An air inlet filter.

Circulation Pump: The circulation pump is installed at the coolant inlet to the FBH unit to assist the coolant flow through the FBH unit and the heater assembly. The pump runs continuously while the FBH unit is in standby or active operating modes. While the FBH unit is inactive, coolant flow is reliant on the engine coolant pump and the auxiliary coolant pump.

Combustion Air Fan: The combustion air fan regulates the flow of air into the unit to support combustion of the fuel supplied by the FBH pump and to purge and cool the FBH unit. A canister type filter is included in the air inlet supply line to prevent particulates entering and contaminating the FBH unit.

Burner Housing: The burner housing contains the burner insert and also incorporates connections for the exhaust pipe, the coolant inlet from the circulation pump and the coolant outlet to the heater assembly. The exhaust pipe directs exhaust combustion gases to atmosphere through a pipe below the FBH unit.

The burner insert incorporates the fuel combustion chamber, an evaporator and a glow plug/flame sensor. Fuel from the FBH fuel pump is supplied to the evaporator, where it evaporates and enters the combustion chamber to mix with air from the combustion air fan. The glow plug/flame sensor provides the ignition source of the fuel:air mixture and, once combustion is established, monitors the flame.

ECU/Heat Exchanger: The ECU controls and monitors operation of the FBH system. Ventilation of the ECU is provided by an internal flow of air from the combustion air fan. The heat exchanger transfers heat generated by combustion to the coolant. A sensor in the heat exchanger provides the ECU with an input of heat exchanger casing temperature, which the ECU relates to coolant temperature and uses to control system operation. The temperature settings in the ECU are calibrated to compensate for the difference between coolant temperature and the heat exchanger casing temperature detected by the sensor. Typically, as the coolant temperature increases, the coolant will be approximately 7 °C (12.6 °F) hotter than the temperature detected by the sensor; as the coolant temperature decreases, the coolant will be approximately 2 °C (3.6 °F) cooler than the temperature detected by the sensor.

Changeover Valve

The changeover valve is a normally open solenoid valve installed between the supply and return sides of the heater coolant circuit. The changeover valve is located in the engine compartment on the engine bulkhead. When de-energised, the changeover valve connects the heater coolant circuit to the engine coolant circuit. When energised, the changeover valve isolates the heater coolant circuit from the engine coolant circuit.

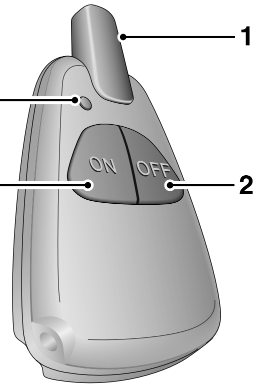
The changeover valve is controlled by a power feed from the ATC ECU.

FBH Receiver

The FBH receiver translates the FBH request radio signals, relayed from the antenna receiver, into a voltage output to the FBH unit. When a request for parked heating is received, the FBH receiver outputs a battery power feed to the FBH unit. When a request to switch off parked heating is received, the FBH receiver disconnects the power feed.

The FBH receiver has a permanent power feed from the vehicle battery and is connected to the antenna receiver by a coaxial cable.

FBH Remote Handset

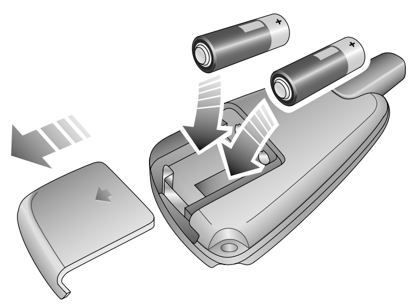


M82 0835

1 Antenna
2 Off button
3 On button
4 LED

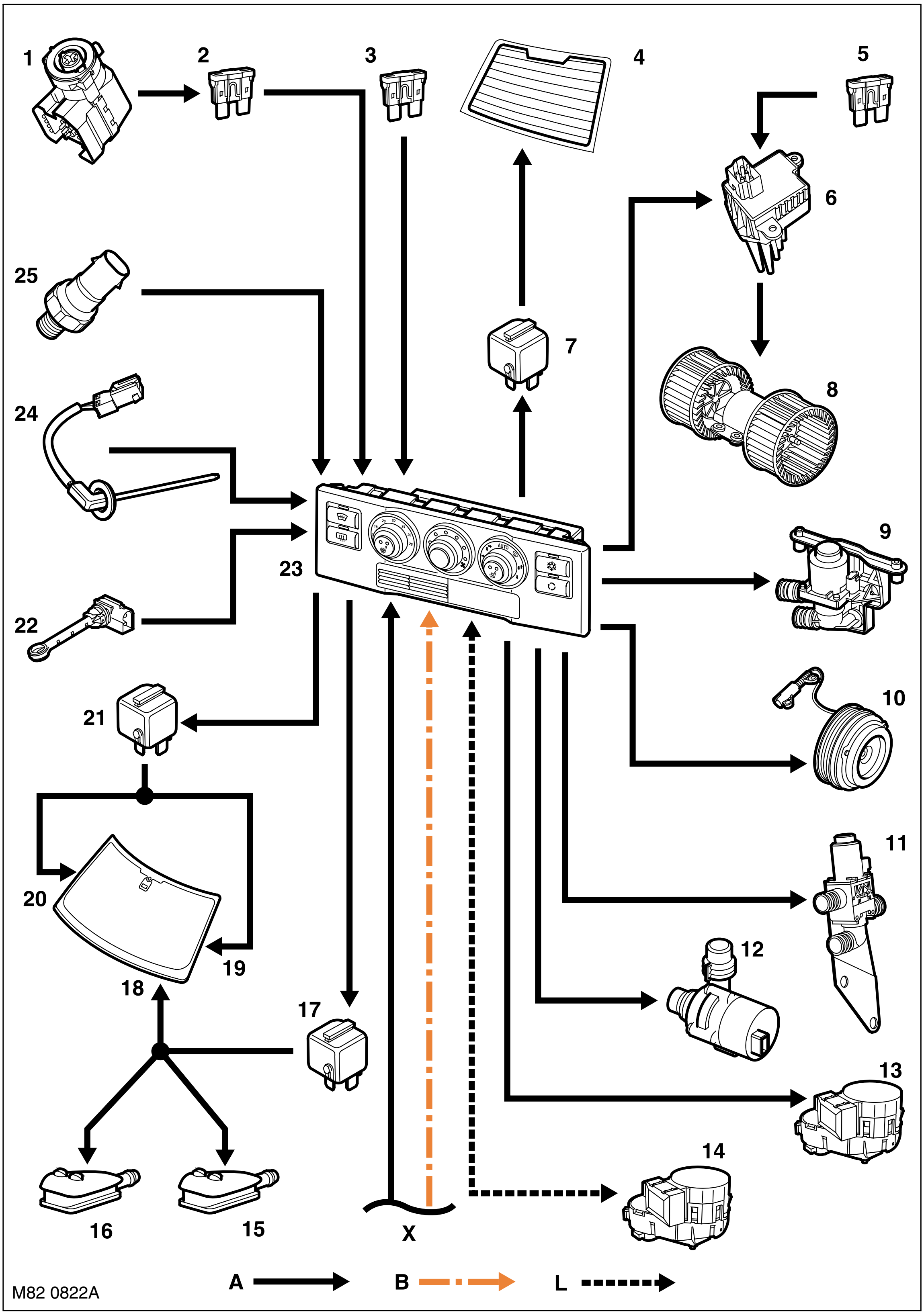
The remote handset allows parked heating to be remotely controlled up to a minimum of 100 m (328 ft) from the vehicle. On and off buttons activate and de-activate parked heating, a red LED illuminates to indicate when parked heating is active and when the on/off signals have been received by the vehicle. The remote handset is powered by two serviceable 1.5 V batteries located under a cover on the rear of the handset.

FBH Remote Handset Battery Replacement



M82 0836

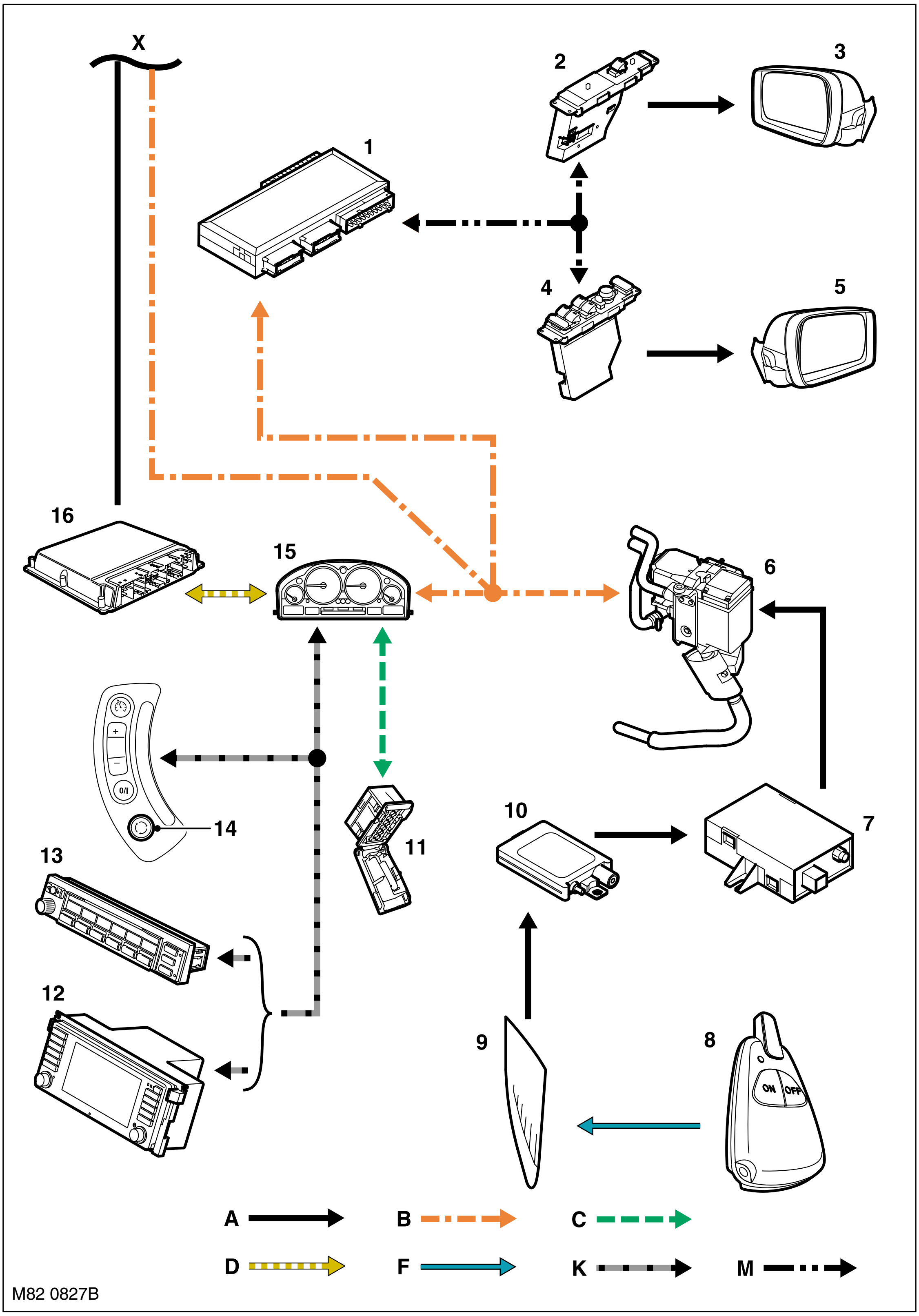
A/C Control Diagram – Low Line System, Sheet 1 of 2



A = Hardwired connection; B = K bus; L = M bus

1 Ignition switch
2 Fuse 34, passenger compartment fusebox
3 Fuse 12, passenger compartment fusebox
4 HRW
5 Fuse 64, passenger compartment fusebox
6 Blower output stage
7 HRW relay
8 Blower
9 Coolant valve
10 Compressor clutch
11 Changeover valve
12 Auxiliary coolant pump
13 Fresh/Recirculated air flaps motor
14 Air distribution motor
15 LH washer jet
16 RH washer jet
17 Washer jet heater relay
18 Wiper park heater
19 Windscreen LH heater
20 Windscreen RH heater
21 Windscreen heater relay
22 Heater temperature sensor
23 ATC ECU
24 Evaporator temperature sensor
25 Refrigerant pressure sensor

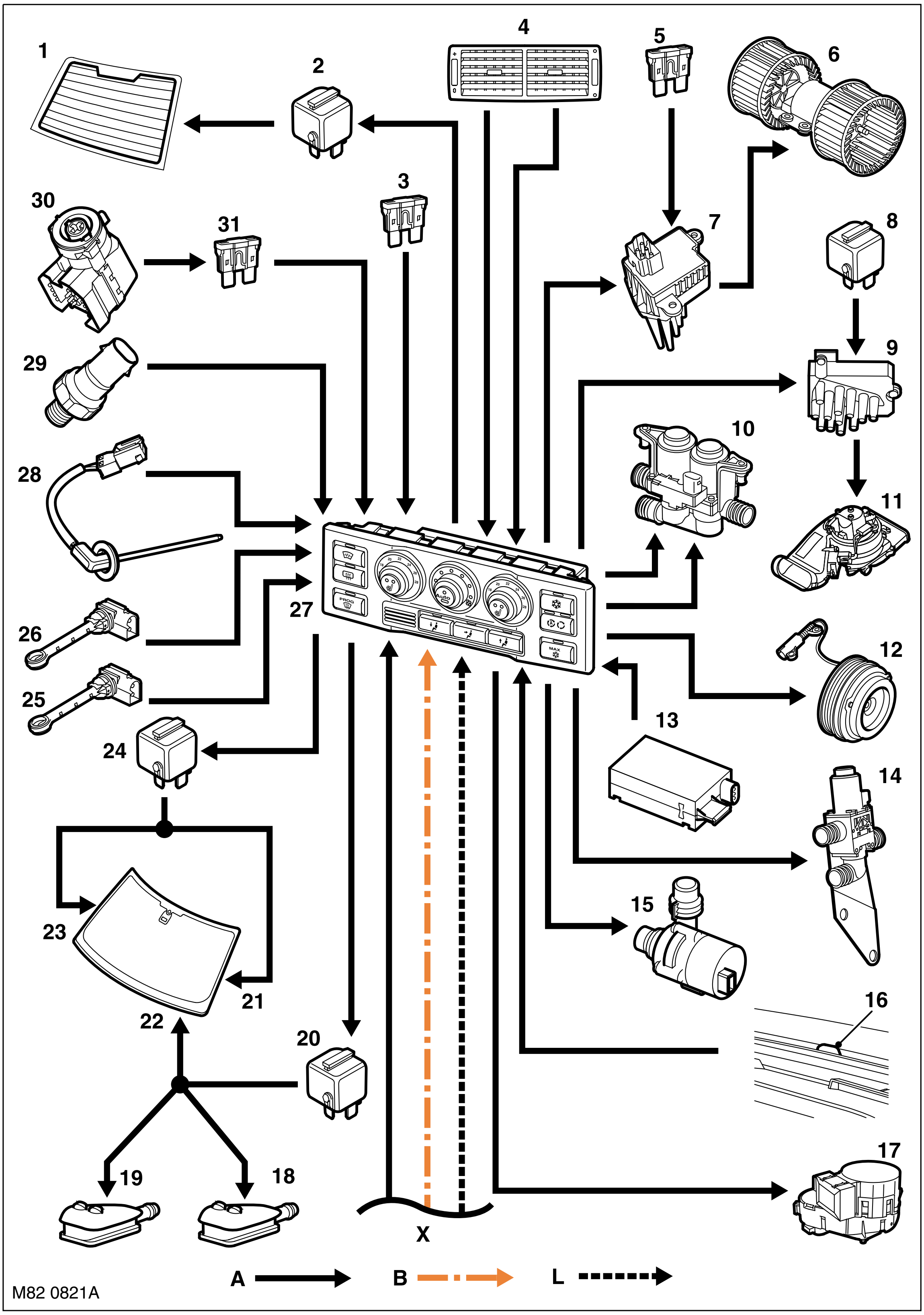
A/C Control Diagram – Low Line System, Sheet 2 of 2



A = Hardwired connection; B = K bus; C = Diagnostic DS2 bus; D = CAN bus; F = RF transmission; K = I bus; M = P bus

1 BCU
2 Passenger door module
3 Passenger door mirror
4 Drivers door module
5 Drivers door mirror
6 FBH unit
7 FBH receiver
8 Remote handset
9 RH side window antenna
10 Antenna amplifier
11 Diagnostic socket
12 Multi function display
13 Multi information display
14 Steering wheel recirculation switch
15 Instrument pack
16 ECM

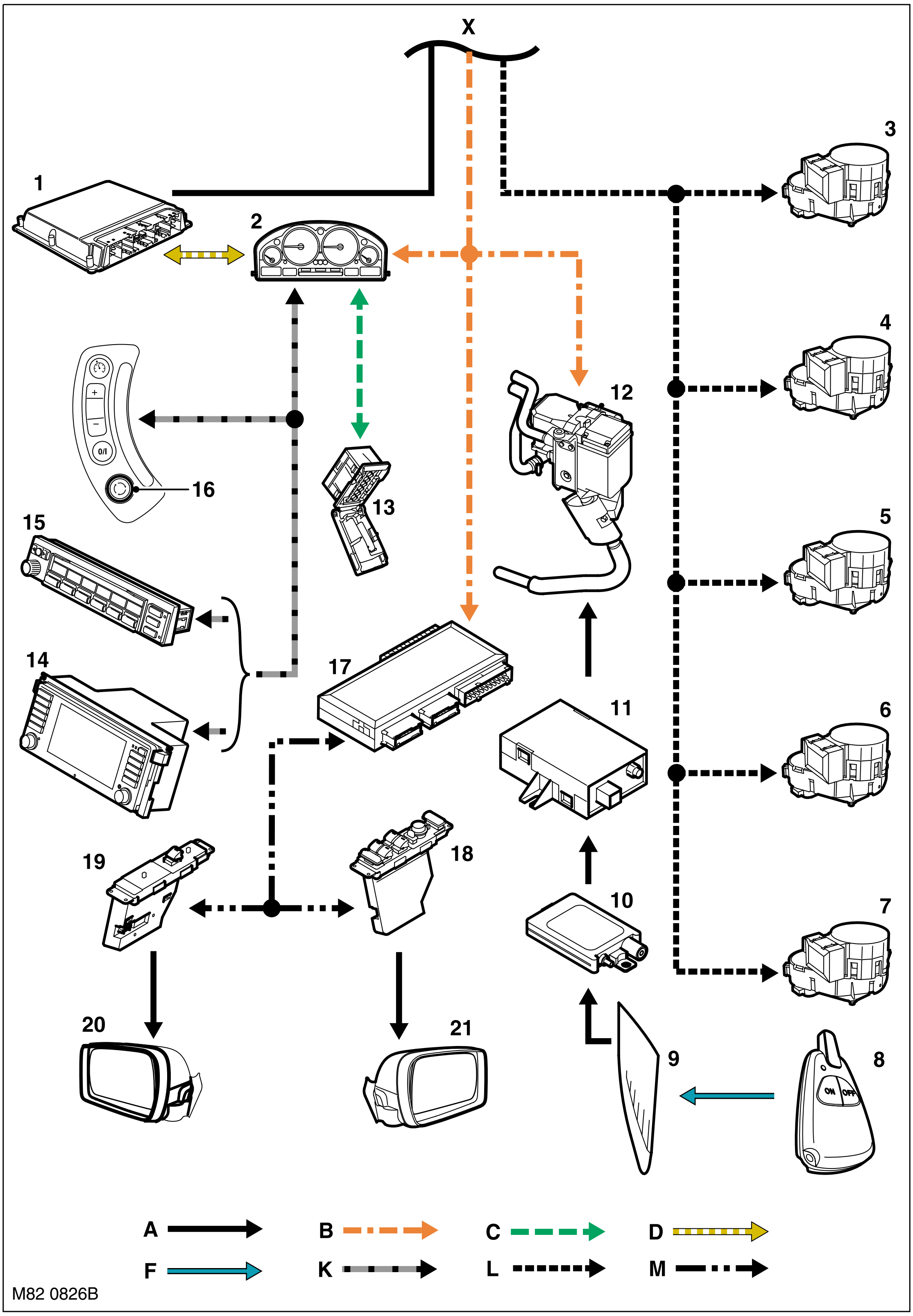
A/C Control Diagram – High Line System, Sheet 1 of 2



A = Hardwired connection; B = K bus; L = M bus

1 HRW
2 HRW relay
3 Fuse 12, passenger compartment fusebox
4 Rear passenger face vent
5 Fuse 64, passenger compartment fusebox
6 Blower
7 Blower output stage
8 Rear blower relay
9 Rear blower output stage
10 Dual coolant valve
11 Rear blower
12 Compressor clutch
13 Pollution sensor
14 Changeover valve
15 Auxiliary coolant pump
16 Sunlight sensor
17 Fresh/Recirculated air flaps motor
18 LH washer jet
19 RH washer jet
20 Washer jet heater relay
21 Wiper park heater
22 Windscreen LH heater
23 Windscreen RH heater
24 Windscreen heater relay
25 RH heater temperature sensor
26 LH heater temperature sensor
27 ATC ECU
28 Evaporator temperature sensor
29 Refrigerant pressure sensor
30 Ignition switch
31 Fuse 34, passenger compartment fusebox

A/C Control Diagram – High Line System, Sheet 2 of 2



A = Hardwired connection; B = K bus; C = Diagnostic DS2 bus; D = CAN bus; F = RF transmission; K = I bus; L = M bus; M = P bus

1 ECM
2 Instrument pack
3 Windscreen distribution motor
4 Face level distribution motor
5 Footwell distribution motor
6 Rear face level temperature blend motor
7 Ram air motor
8 Remote handset
9 RH side window antenna
10 Antenna amplifier
11 FBH receiver
12 FBH unit
13 Diagnostic socket
14 Multi function display
15 Multi information display
16 Steering wheel recirculation switch
17 BCU
18 Drivers door module
19 Passenger door module
20 Passenger door mirror
21 Drivers door mirror

Operation

General

Both the low and high line systems operate on the reheat principle. The air entering the heater assembly is cooled to a constant value by the evaporator and then reheated as necessary by the heater matrix to produce the temperature(s) selected on the control panel.

To determine the various system settings, the ATC ECU derives a reference value (called the Y factor) from:

- The temperature setting on the control panel
- The ambient temperature
- The in-car temperature.

The reference value is measured in %, where –27.5% means maximum cooling is required and 100% means maximum heating is required. On the high line system, separate reference values are produced for the LH and RH sides of the heater assembly.

On both the low and high line systems the reference value is used for temperature control. On the high line system the driver's side reference value is also used for flap positioning and blower speed calculations.

When the ignition is turned off the ATC ECU memorises the system settings and resumes the same settings the next time the ignition is switched on.

Compressor Control

The compressor is engaged by pressing either the automatic mode switch, defrost switch, A/C switch or maximum A/C switch. To prevent a dip in engine speed when the engine is at idle, a time delay of approximately 0.5 second is built into the compressor engagement process. The time delay allows the ECM to increase throttle angle and fuelling in anticipation of the additional load on the engine when the compressor engages.

When it receives an input to engage the compressor, the ATC ECU sends a message to the ECM, via the K bus, instrument pack and CAN bus, to advise that it wants to engage the compressor. Provided there are no engine management problems, the ECM responds by increasing throttle angle and fuelling and sending a message granting the request to the ATC ECU over the CAN bus, instrument pack and K bus. When it receives the grant message, the ATC ECU energises the compressor clutch provided the following conditions exist:

- Engine speed is more than 400 rev/min
- Evaporator temperature is more than 3 °C (37 °F)
- The refrigerant pressure is within limits
- Battery voltage is less than 16 V
- The blower is running
- There are no faults detected by the ATC ECU.

The compressor remains engaged until selected off or the required conditions no longer exist. If the evaporator temperature decreases to approximately 2 °C (36 °F) the compressor is disengaged, then re-engaged when the evaporator temperature increases to more than 3 °C (37 °F) again. If battery voltage exceeds 16 V for more than 5 seconds the compressor is disengaged, then re-engaged when voltage decreases to less than 15.8 V.

The compressor can also be disengaged by the ECM changing the grant message to a negative value or outputting a hardwired signal direct to the ATC ECU. Changing the grant message involves a time delay of up to 4 seconds before the ATC ECU de-energises the compressor clutch. The time delay allows the ECM to decrease throttle angle and fuelling, in anticipation of the reduction in engine load when the compressor disengages, to prevent a sudden increase in engine speed if the engine is at idle. The hardwired signal is used to obtain instant disengagement of the compressor to ensure maximum acceleration when accelerator pedal demand suddenly goes from no load to full load. When this occurs the ECM sends a 12 V signal on a hardwired connection direct to the ATC ECU. When it receives the signal the ATC ECU immediately de-energises the compressor clutch.